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REEL SETS AND SYMBOLS FOR AWARDS****Publication Classification**(51) **Int. Cl.**
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17/3286 (2013.01)(71) Applicant: **IGT, Las Vegas, NV (US)**(72) Inventor: **Adam Narducci, Reno, NV (US)**(21) Appl. No.: **19/213,189**(22) Filed: **May 20, 2025****Related U.S. Application Data**

(63) Continuation of application No. 19/212,170, filed on May 19, 2025, which is a continuation of application No. 17/939,644, filed on Sep. 7, 2022, now Pat. No. 12,333,891.

(57) **ABSTRACT**

Gaming systems and methods that activate reel sets responsive to accumulations of points and provides awards responsive to accumulations of symbols with any activated reel sets.

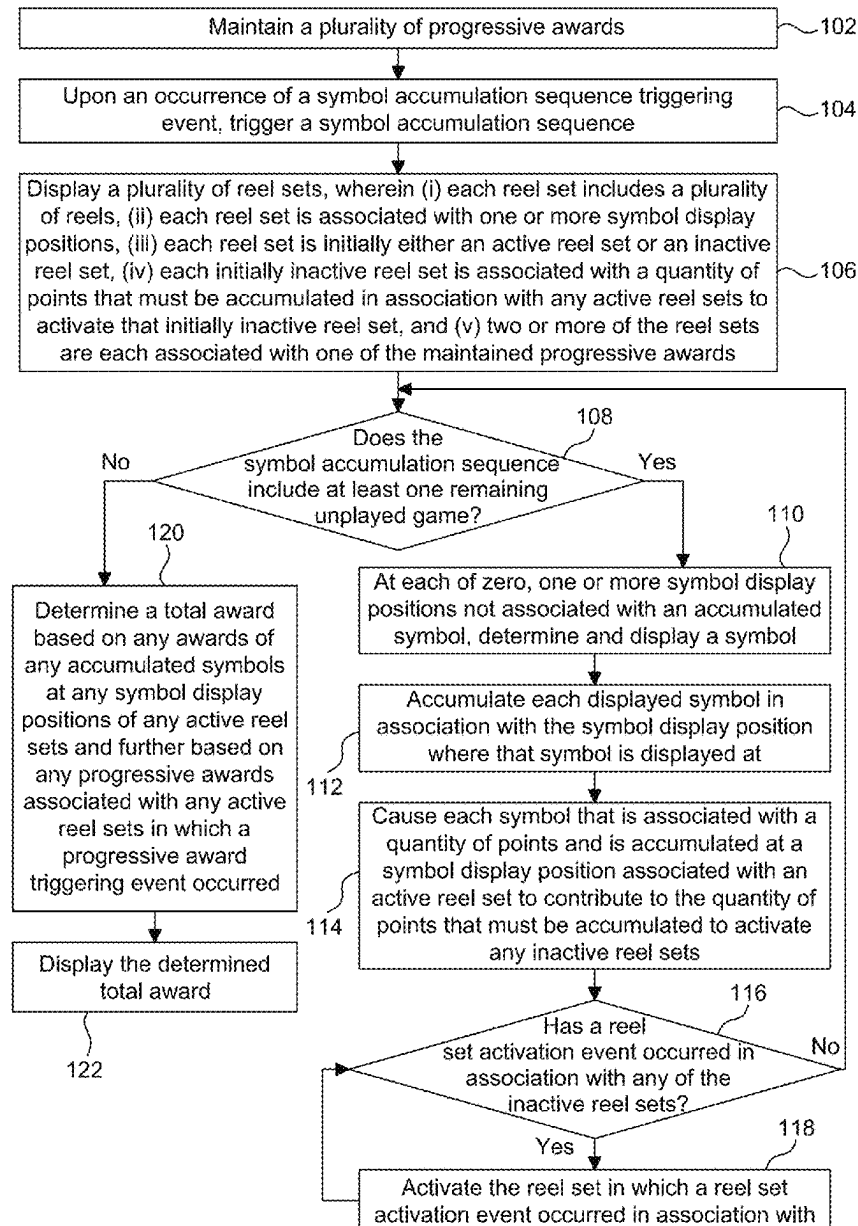


FIG. 1

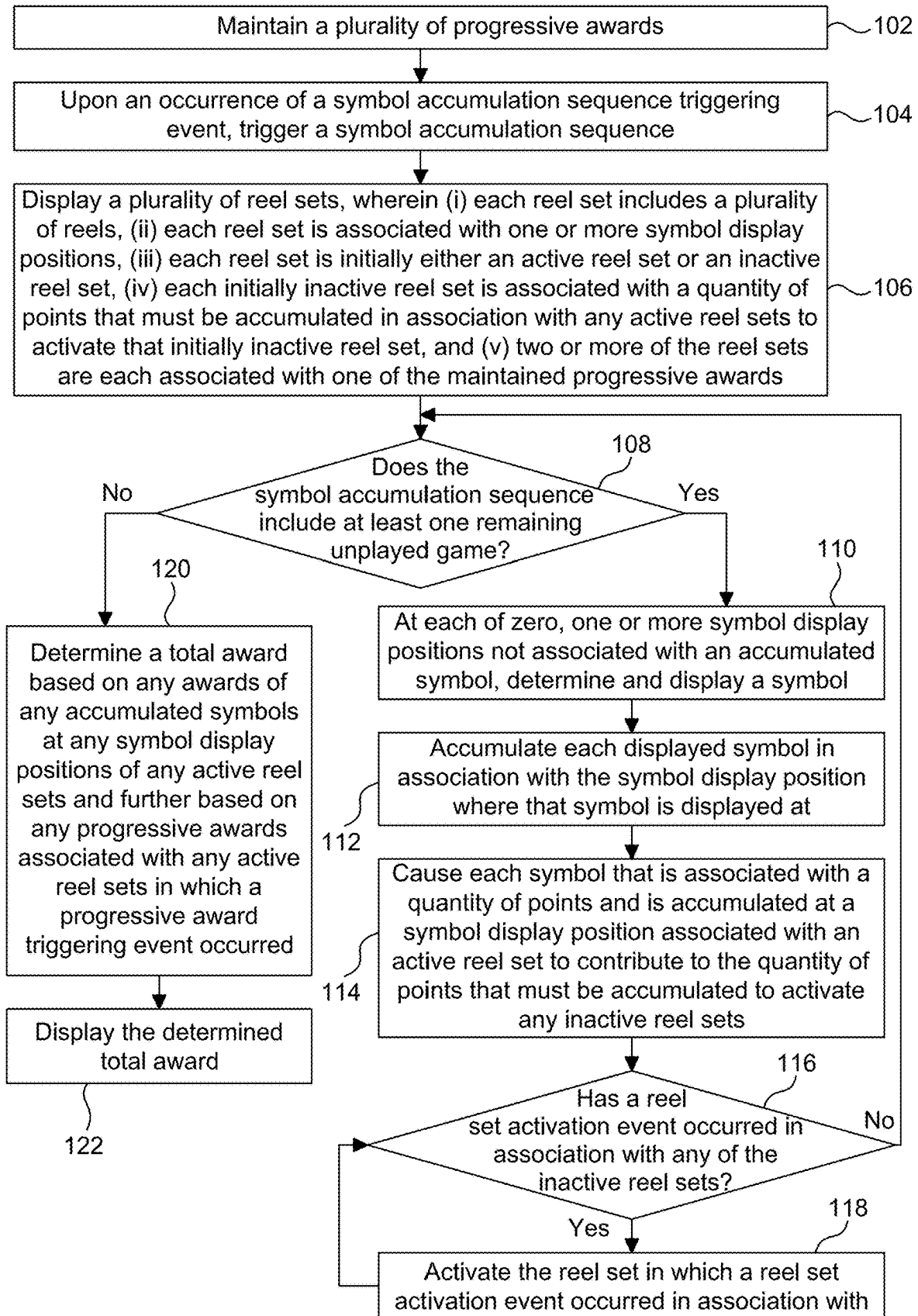


FIG. 2A

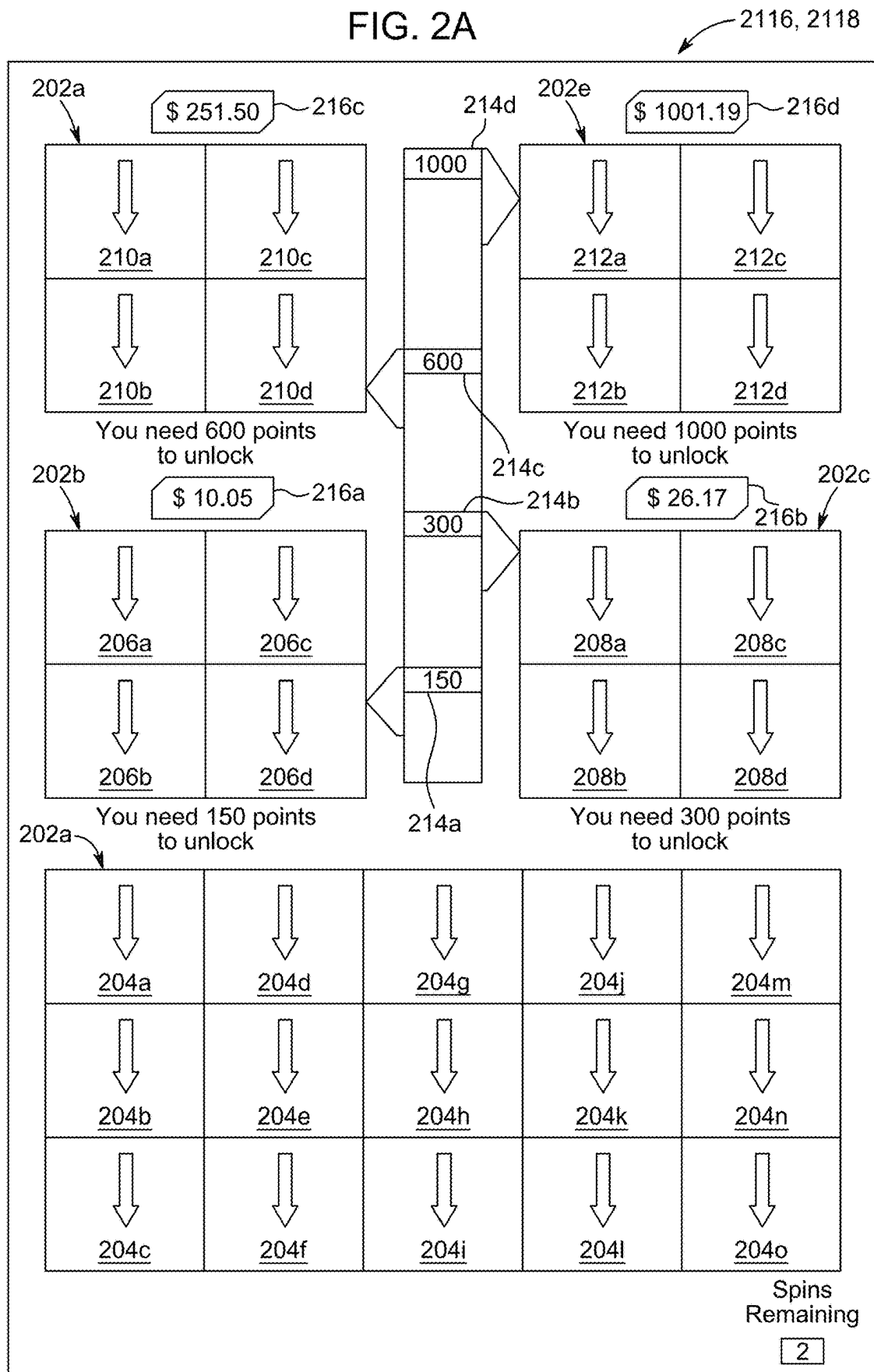


FIG. 2B

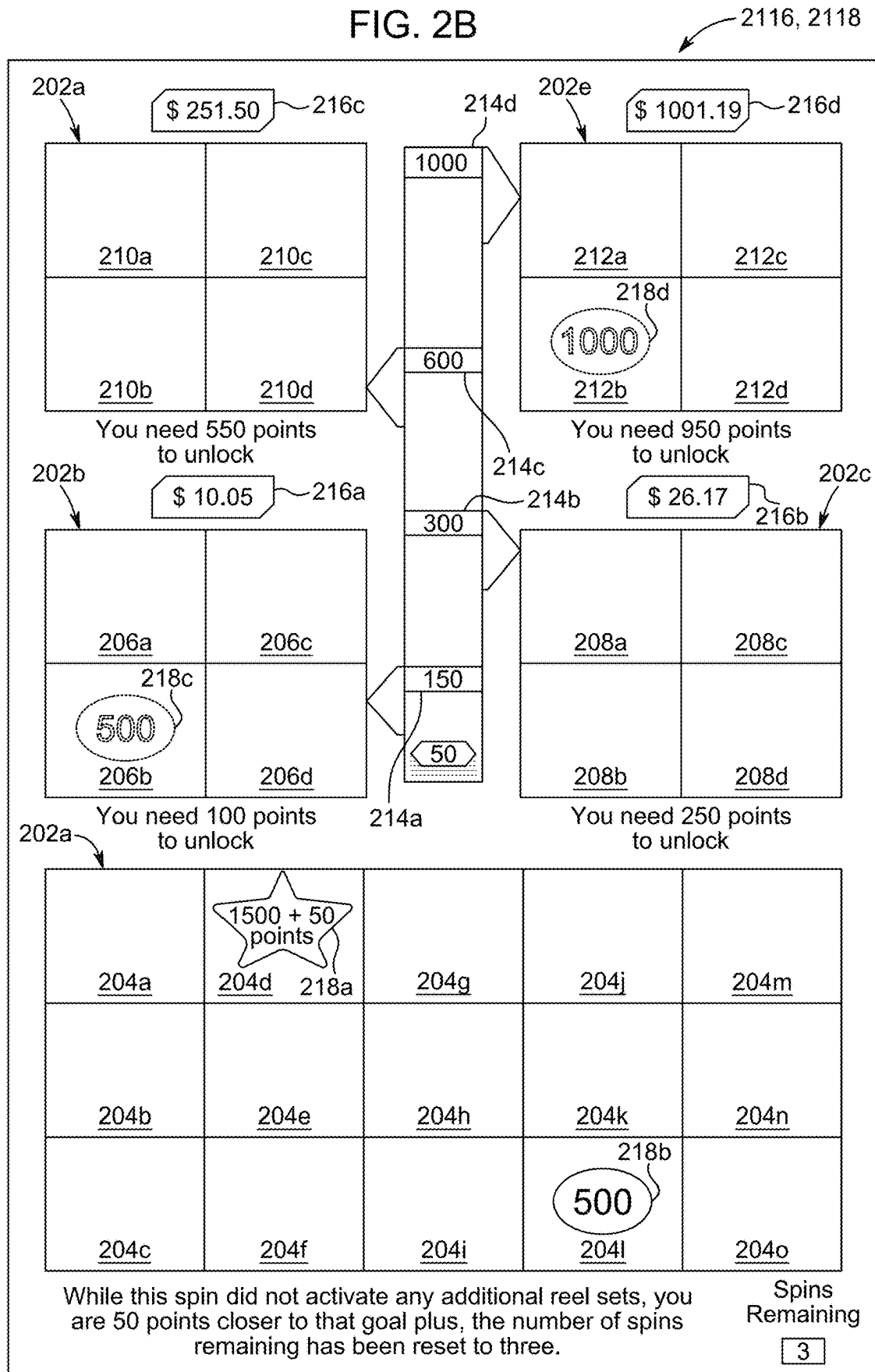


FIG. 2C

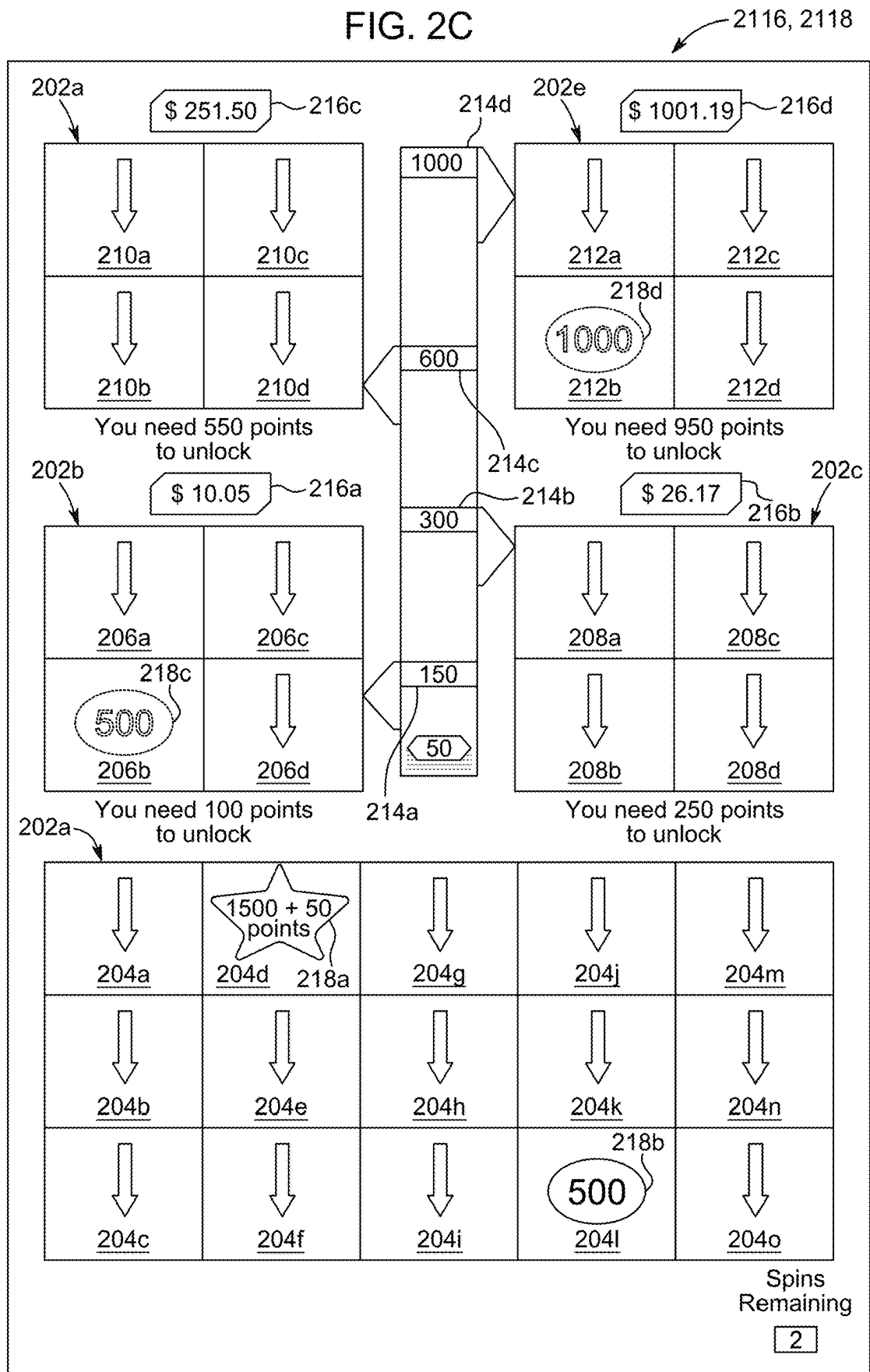


FIG. 2D

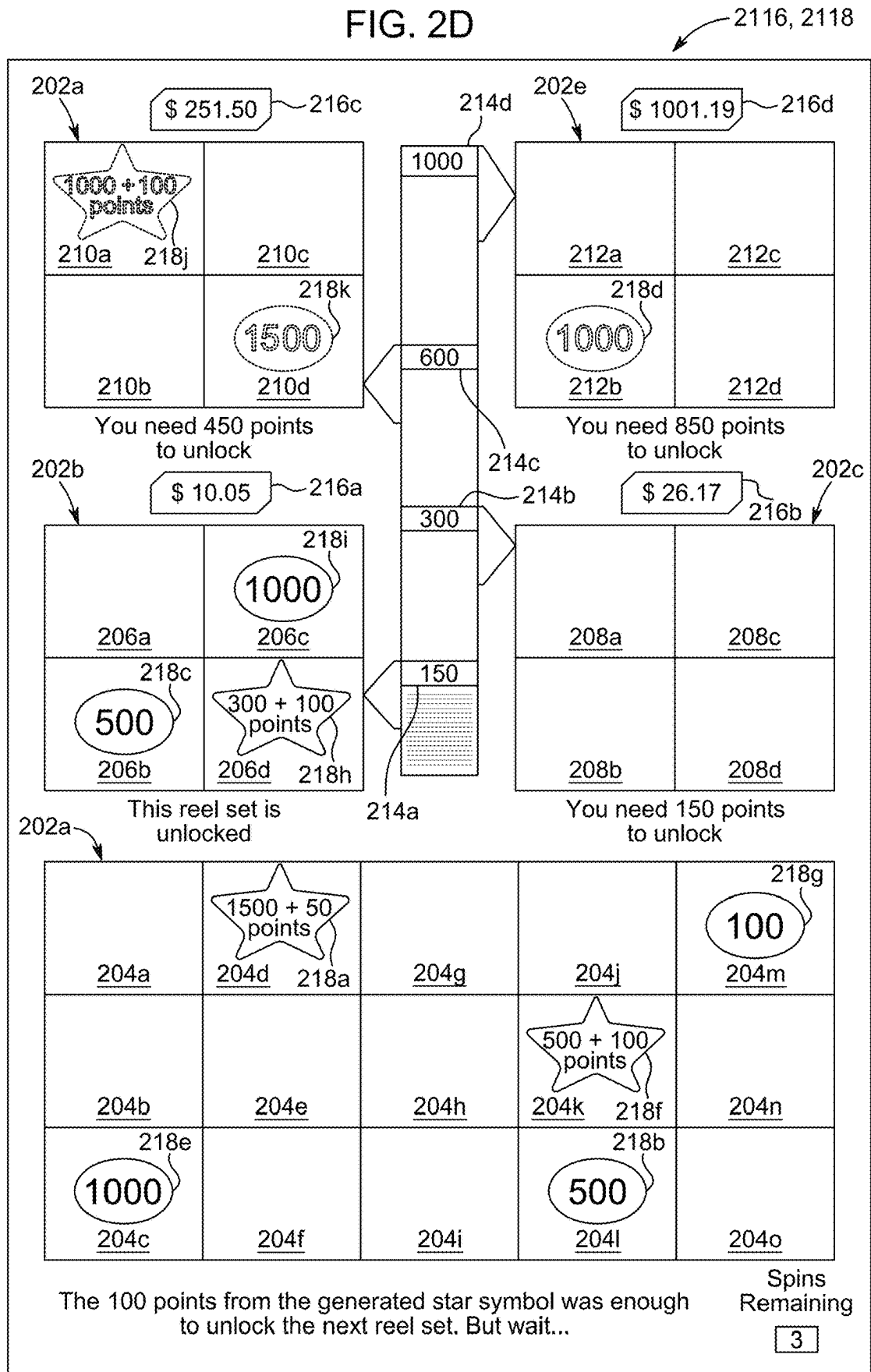


FIG. 2E

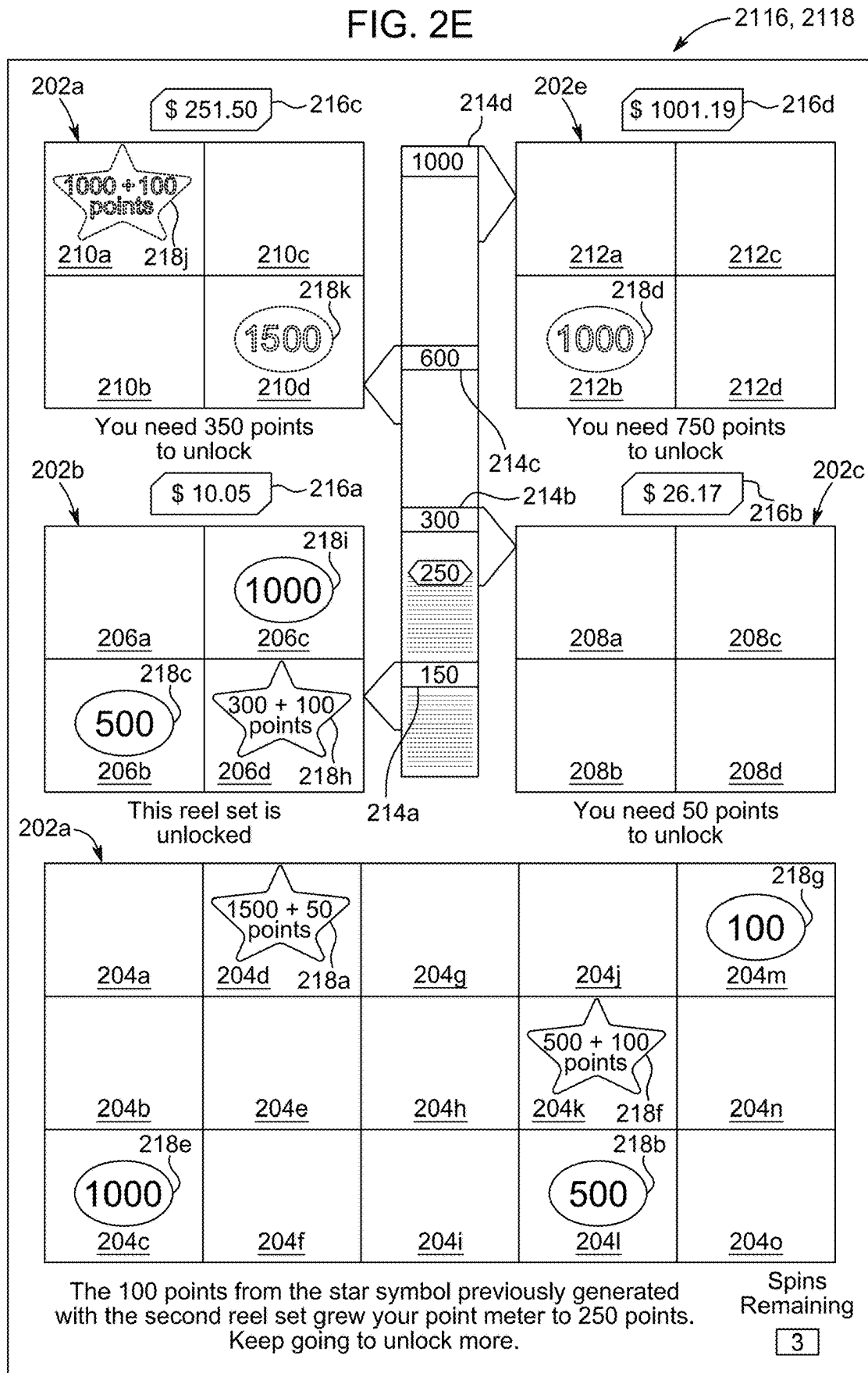
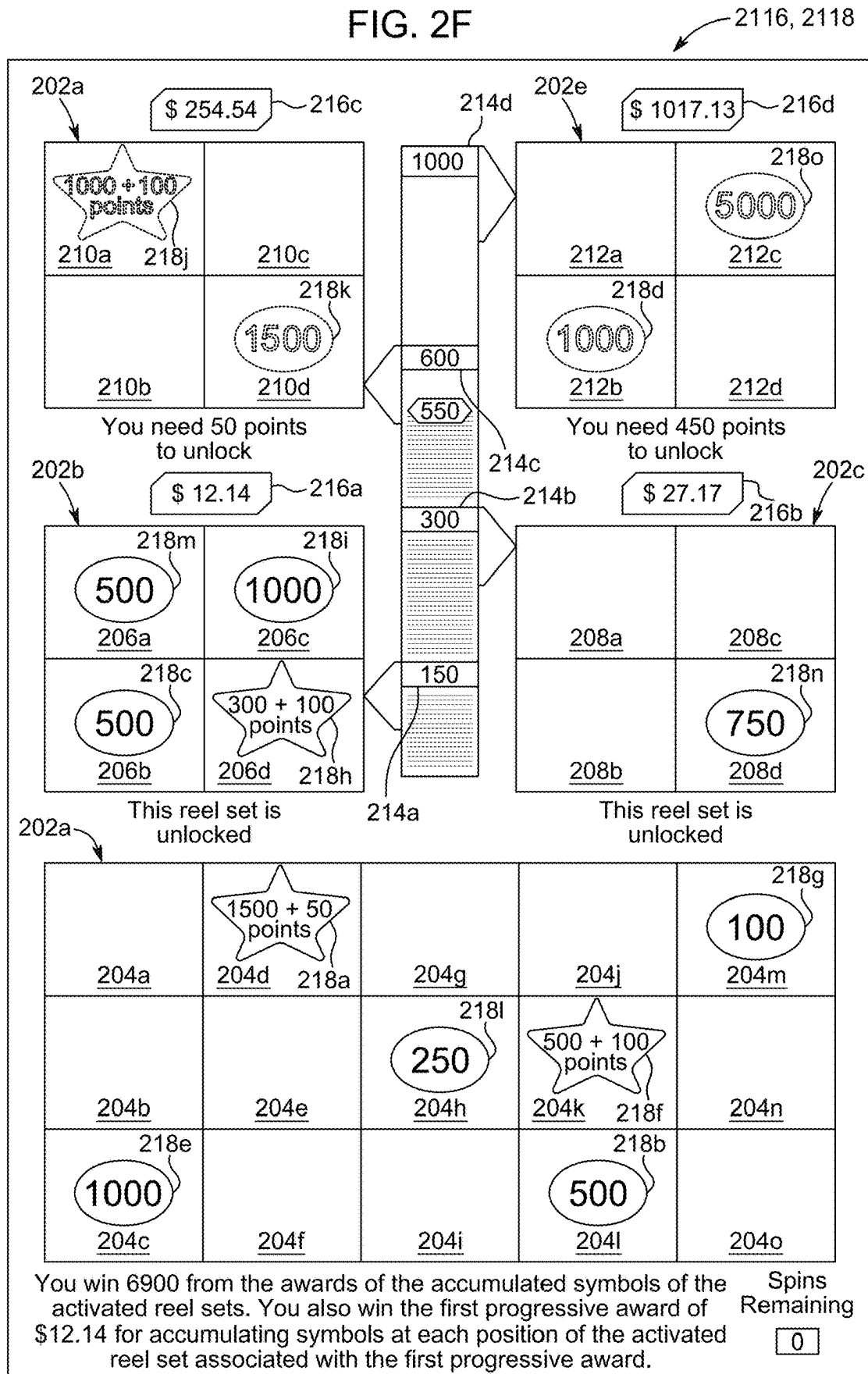


FIG. 2F



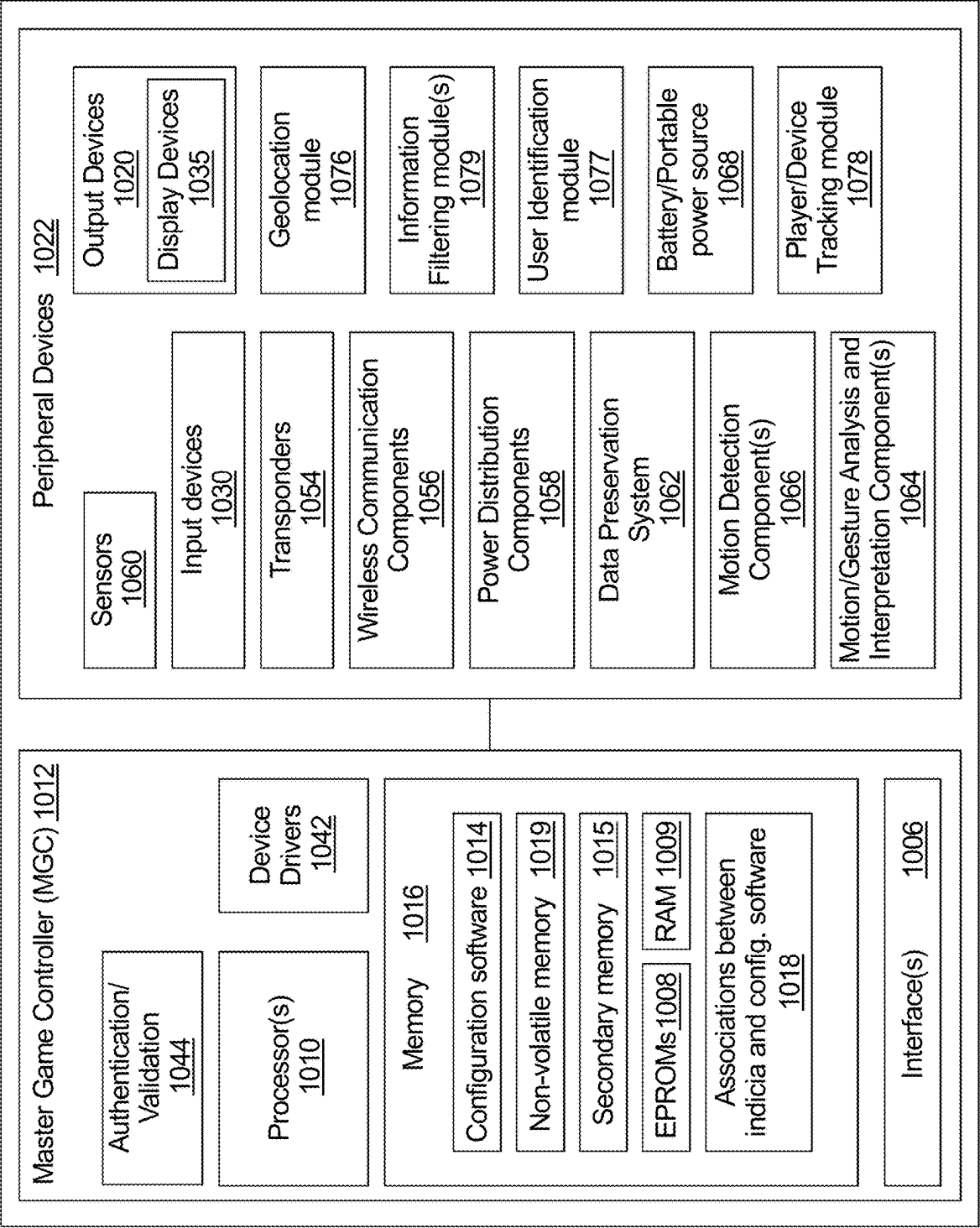


FIG. 4A

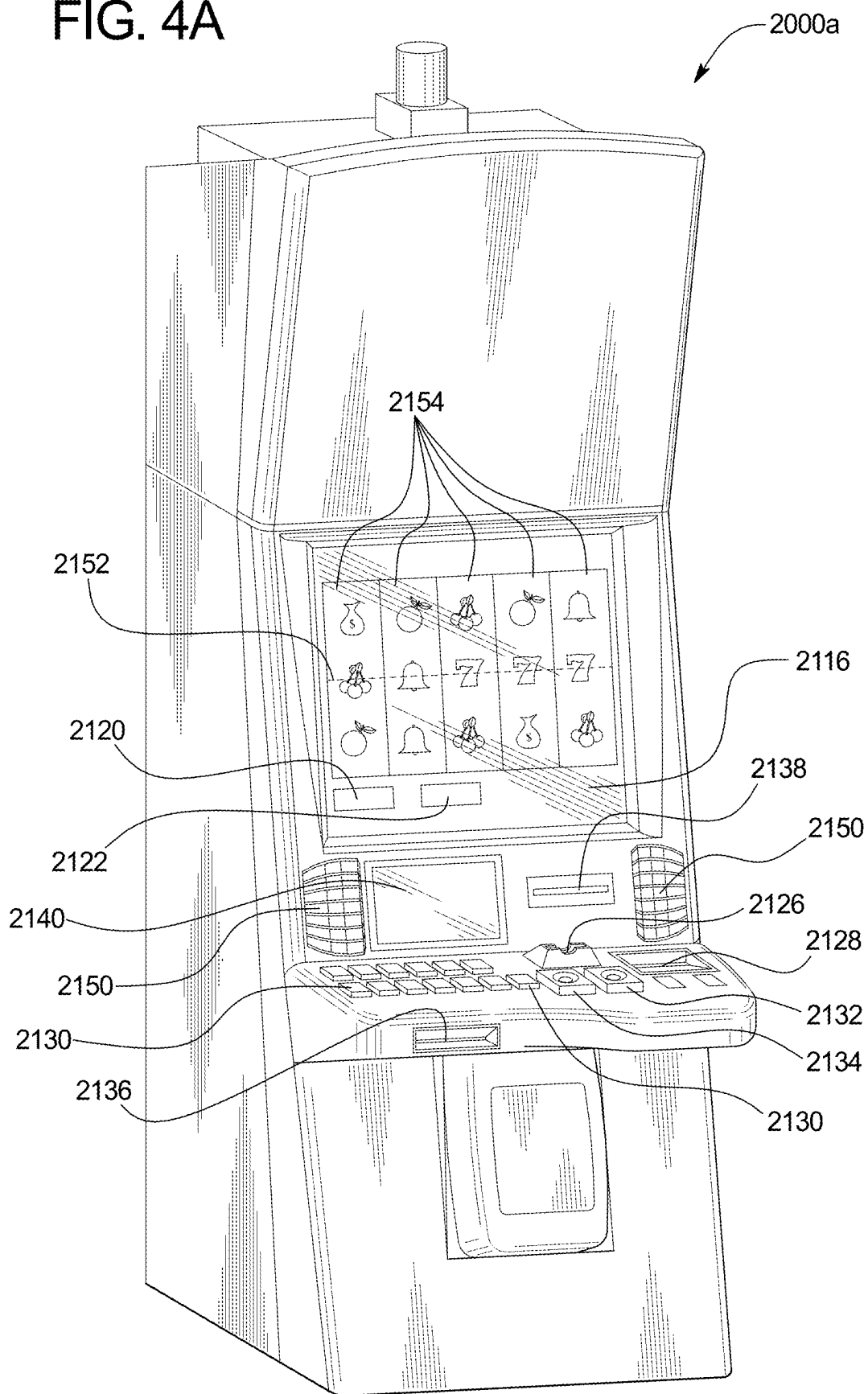


FIG. 4B

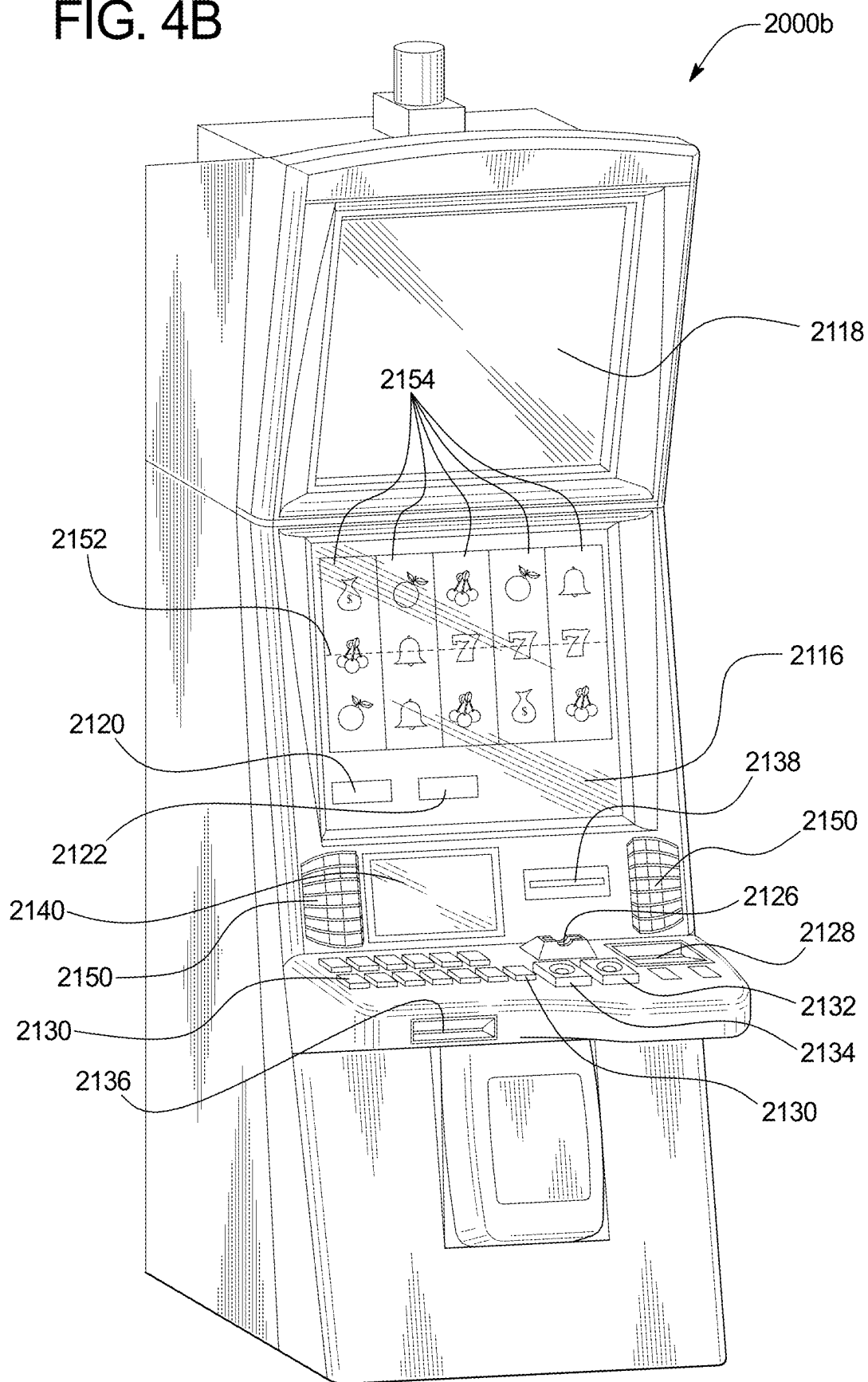
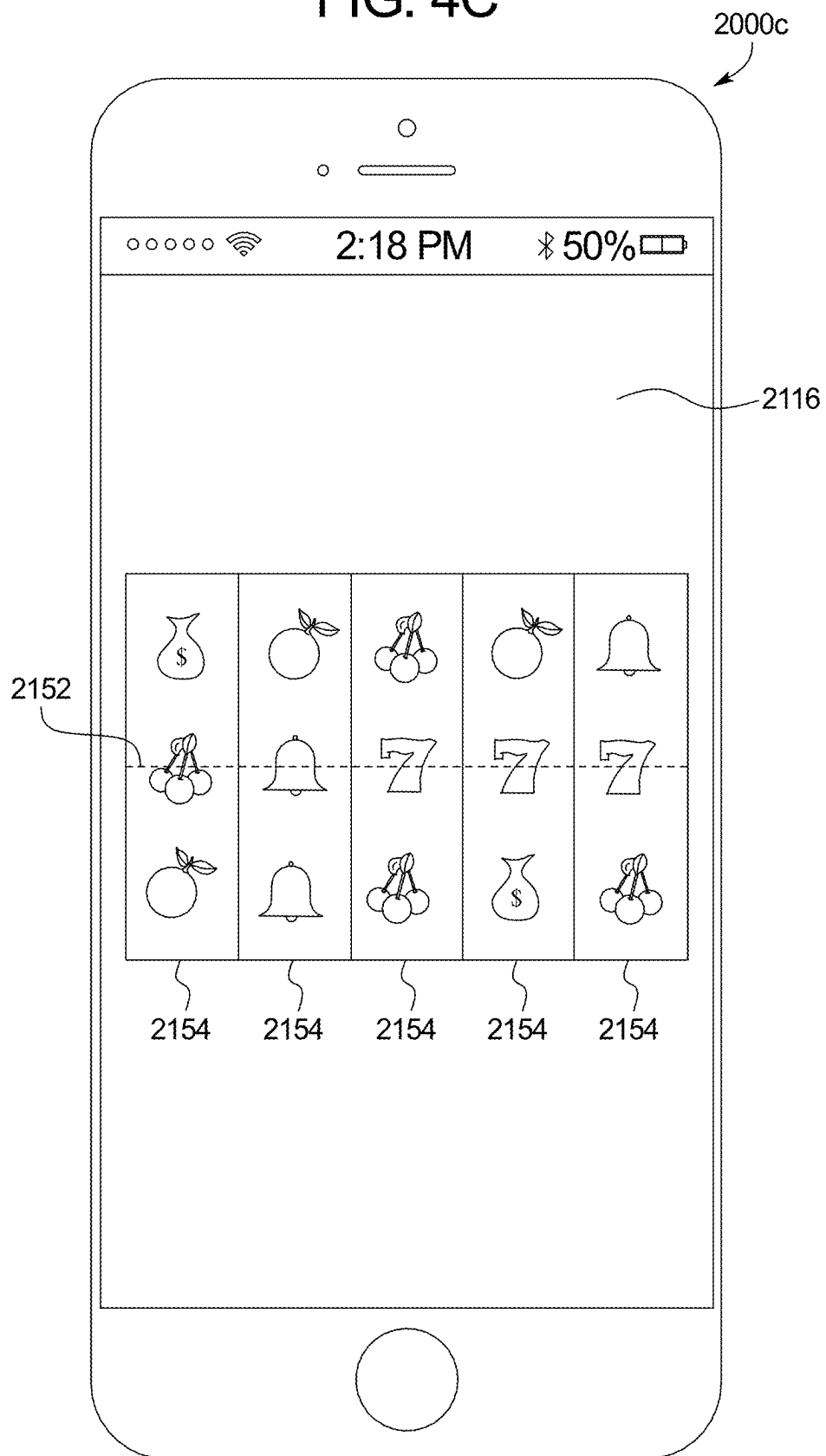


FIG. 4C



ACCUMULATING POINTS TO ACTIVATE REEL SETS AND SYMBOLS FOR AWARDS

PRIORITY CLAIM

[0001] This application is a continuation of, claims the benefit of and priority to U.S. patent application Ser. No. 19/212,170, filed on May 19, 2025, which is a continuation of, claims the benefit of and priority to U.S. patent application Ser. No. 17/939,644, filed on Sep. 7, 2022, the entire contents of which are each incorporated herein by reference.

BACKGROUND

[0002] In various embodiments, the systems and methods of the present disclosure activate reel sets responsive to accumulations of points and provides awards responsive to accumulations of symbols with any activated reel sets.

[0003] Gaming machines may provide players awards in primary games. Gaming machines generally require the player to place or make a wager to activate the primary or base game. The award may be based on the player obtaining a winning symbol or symbol combination and on the amount of the wager.

BRIEF SUMMARY

[0004] In certain embodiments, the present disclosure relates to a gaming system including a processor, and a memory device that stores a plurality of instructions. When executed by the processor in association with a sequence including a plurality of plays of a game and responsive to less than a first quantity of points accumulated, over the plurality of plays of the game, in association with symbols displayed at a plurality of symbol display positions associated with a first reel set including a first plurality of displayed reels, the instructions cause the processor to determine a first total award including any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the first reel set, wherein the first total award is determined independent of any awards associated with any symbols accumulated at any of a plurality of symbol display positions associated with a second reel set including a second plurality of displayed reels. In these embodiments, when executed by the processor responsive to at least the first quantity of points accumulated, over the plurality of plays of the game, in association with symbols displayed at the plurality of symbol display positions associated with the first reel set, the instructions cause the processor to determine a second total award including any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the first reel set and any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the second reel set, and communicate data that results in a display, by a display device, of any determined total award.

[0005] In certain embodiments, the present disclosure relates to a gaming system including a processor, and a memory device that stores a plurality of instructions. When executed by the processor in association with a sequence including a plurality of plays of a game, the instructions cause the processor to communicate data that results in a display, by a display device, of a plurality of reel sets, wherein the plurality of reel sets includes a first reel set in an active state, and a second reel set in an inactive state. In

these embodiments, when executed by the processor responsive to an occurrence of an accumulation of a first quantity of points associated with the second reel set, the instructions cause the processor to modify the second reel set from being in the inactive state to being in the active state. Moreover, when executed by the processor responsive to an occurrence of a sequence award determination event, the instructions cause the processor to determine a total award, the determination being based on any accumulation of any symbols, over the plurality of plays of the game, at any symbol display positions associated with any of the plurality of reel sets in the active state when the sequence award determination event occurs, and communicate data that results in a display, by the display device, of the determined total award.

[0006] In certain embodiments, the present disclosure relates to a gaming system including a processor, and a memory device that stores a plurality of instructions. When executed by the processor in association with a sequence including a plurality of plays of a game, the instructions cause the processor to communicate data that results in a display, by a display device, of a plurality of reel sets, wherein: each reel set includes a plurality of symbol display positions, and when the sequence is triggered, each reel set is one of an active reel set and an inactive reel set. In these embodiments, when executed by the processor, the instructions cause the processor to communicate data that results in a display, by the display device, of an accumulation of symbols, over the plurality of plays of the game, at a quantity of the symbol display positions associated with any of the active reel sets and any of the inactive reel sets, and responsive to an accumulation of a quantity of points associated with any of the inactive reel sets, activate that reel set. In such embodiments, when executed by the processor responsive to an occurrence of sequence award determination event, the instructions cause the processor to determine a total award, wherein the total award is based on any symbols accumulated at any of the symbol display positions associated with any of the active reel sets and is independent of any symbols accumulated at any of the symbol display positions associated with any of the inactive reel sets, and communicate data that results in a display, by the display device, of the determined total award.

[0007] Additional features are described herein, and will be apparent from the following Detailed Description and the figures.

BRIEF DESCRIPTION OF THE FIGURES

[0008] FIG. 1 is a flow chart an example process for operating a gaming system that accumulates points to activate reel sets and accumulates symbols to determine a total award.

[0009] FIGS. 2A, 2B, 2C, 2D, 2E, and 2F are front views of one embodiment of the gaming system of the present disclosure illustrating a play of a sequence including the accumulation of points to activate reel sets and the accumulation of symbols to determine a total award.

[0010] FIG. 3 is a schematic block diagram of one embodiment of an electronic configuration of an example electronic gaming machine of the present disclosure.

[0011] FIGS. 4A and 4B are perspective views of example alternative embodiments of an electronic gaming machine of the present disclosure.

[0012] FIG. 4C is a front view of an example personal gaming device of the present disclosure.

DETAILED DESCRIPTION

[0013] In various embodiments, the present disclosure relates generally to gaming systems and methods that activate reel sets responsive to accumulations of points and provides awards responsive to accumulations of symbols with any activated reel sets.

[0014] In certain embodiments, responsive to an occurrence of a sequence triggering event, the gaming system displays a plurality of reel sets. Each reel set includes a plurality of reels associated with a plurality of symbol display positions. One or more of the reel sets are initially active and zero, one or more of the reel sets are initially inactive. Each initially inactive reel set is associated with a quantity of points that must be accumulated to activate that inactive reel set. For example, upon the triggering of the sequence, the gaming system activates a first reel set without activating a second reel set (which is activated upon a first quantity of points being accumulated in association with symbols displayed at symbol display positions associated with active reel sets) or activating a third reel set (which is activated upon a second quantity of points being accumulated in association with symbols displayed at symbol display positions associated with active reel sets).

[0015] In certain embodiments, in addition to including a plurality of symbol display positions, certain reel sets are each associated with an award, such as a different ranked progressive award of a multi-level progressive award configuration. In these embodiments, any awards associated with any initially active reel sets are initially active awards and any awards associated with any initially inactive reel sets are initially inactive awards. Continuing with the above example, if the gaming system activates a first reel set without activating any of the remaining reel sets, then the progressive award associated with that first reel set is initially active (i.e., may be provided in association with the triggered sequence) while the remaining progressive awards associated with the remaining reel sets are initially inactive (i.e., may not be provided in association with the triggered sequence unless otherwise activated).

[0016] In operation of certain embodiments, for each play of a game of the triggered sequence, the gaming system randomly determines and displays zero, one or more symbols at zero, one or more available symbol display positions of zero, one or more of the displayed plurality of reel sets. For each randomly determined and displayed symbol at each symbol display position of each of the reel sets, the gaming system accumulates that symbol in association with that symbol display position. In these embodiments, the gaming system operates to display and accumulate zero, one or more symbols at zero, one or more symbol display positions associated with any active reel sets and/or any inactive reel sets. That is, a symbol display position associated with an active reel set or a symbol display position associated with an inactive reel set may each accumulate a symbol in association with the triggered sequence. It should be appreciated that once accumulated in association with a symbol display position, that accumulated symbol remains associated with that symbol display position for the duration of the triggered sequence such that a symbol display position associated with an accumulated symbol (or a quantity of accumulated symbols) from one play of the game of the triggered sequence is unavailable to display another symbol for another play of the game of the triggered sequence.

[0017] In certain embodiments, each accumulated symbol is associated with a displayed award. In certain such embodiments, the accumulation of a symbol associated with an award at a symbol display position of an active reel set results in that award being included in a total award determined at the conclusion of the triggered sequence, while the accumulation of a symbol associated with an award at a symbol display position of an inactive reel set results in that award not being included in a total award determined at the conclusion of the triggered sequence. As such, an accumulated symbol associated with an award may or may not result in that award being provided based on whether or not the symbol display position where that symbol is displayed at is associated with an active reel set and/or an inactive reel set when the sequence concludes. Such a configuration provides that certain awards associated with certain symbols may be displayed and accumulated as part of the triggered sequence but not necessarily provided during the triggered sequence. This accumulation of symbols associated with awards without such awards being provided increase the volatility of the operation of the gaming system and triggers a near-miss reaction in certain users that increases the level of anticipation such user's experience.

[0018] In addition to determining, displaying and accumulating symbols at zero, one or more symbol display positions associated with zero, one or more reel sets (regardless of the state of each reel set being in an active state or an inactive state), the gaming system determines if any points determined in association with any symbols displayed at any symbol display positions associated with any active reel sets cause the activation of any inactive reel sets. In these embodiments, since each initially inactive reel set is associated with a quantity of points need to activate that inactive reel set, the gaming system determines whether or not the quantity of points associated with each remaining inactive reel set has been accumulated (based on certain symbols displayed at symbol display positions associated with active reel sets) and if so, causes a modification of the state of that reel set from an inactive state to an active state. Continuing with the above example, following the determination, display and accumulation of zero, one or more symbols at zero, one or more symbol display positions associated with zero, one or more reel sets (and assuming that the first reel set remains the only active reel set) and following the determination of a quantity of points associated with any designated symbols accumulated at any of the symbol display positions associated with that first active reel set, the gaming system determines if the first quantity of points associated with the inactive second reel set have been earned (and if so, causes the activation of the second reel set) and/or if at least the second quantity of points associated with the inactive third reel set have been earned (and if so, causes the activation of the third reel set).

[0019] This accumulation of zero, one or more symbols at zero, one or more symbol display positions over one or more plays of the game of the triggered sequence and this activation of zero, one or more reel sets based on points accumulated continues until the gaming system determines that a sequence termination event has occurred. Upon the occurrence of the sequence termination event, such as upon the gaming system determining that no unplayed games remain associated with the triggered sequence, the gaming system determines a total award for the triggered sequence. In certain embodiments, the total award includes any awards

associated with any accumulated symbols at any symbol display positions associated with any active reel sets, but does not include any awards associated with any accumulated symbols at any symbol display positions associated with any inactive reel sets. In the certain embodiments, the total award additionally or alternatively includes zero, one or more awards associated with activated reel sets. In these embodiments, the gaming system determines whether or not to include any awards, such as progressive awards associated with specific reel sets, based on if a symbol is accumulated at a designated quantity of the symbol display positions of the active reel set associated with that award. Such a configuration provides a change in operation of the gaming system (relative to prior gaming systems) whereby the activation of a reel set activates an award to potentially be won but the accumulation of symbols at symbol display positions of that activated reel set determine whether or not the activated award is provided as part of the total award. Such additional determinations alter the volatility of the gaming system resulting in a gaming system that operates differently in providing zero, one or more awards.

[0020] It should be appreciated that in certain embodiments, while the accumulation of a symbol associated with an award at a symbol display position associated with an inactive reel set may not initially be part of any determined total award when that symbol is accumulated, the award associated with that symbol may subsequently be part of any determined total award if the reel set associated with the symbol display position where that symbol is accumulated subsequently becomes an active reel set during the triggered sequence. Similarly, in certain embodiments, while a quantity of points associated with a symbol displayed at a symbol display position associated with an inactive reel set may not initially be employed to activate any additional reel sets, such a quantity of points associated with that symbol may subsequently be employed as part of an activation of one or more additional reel sets if the reel set associated with the symbol display position where that symbol is displayed subsequently becomes an active reel set during the triggered sequence. As such, an award associated with a symbol displayed at a symbol display position associated with an inactive reel set and/or a quantity of points associated with a symbol displayed at a symbol display position associated with an inactive reel set may change from not being available to being available based on one or more events (i.e., the potential activation of the inactive reel set) occurring during the triggered sequence. Such a configuration provides an additional change in operation of the gaming system (relative to prior gaming systems) whereby certain awards and/or points associated with certain symbols are determined and displayed during the triggered sequence but such awards and/or points may or may not ultimately be provided as part of the triggered sequence.

[0021] While certain embodiments described below are directed to accumulating symbols and activating reel sets in association with one or more plays of a secondary game, such as a bonus game, it should be appreciated that such embodiments may additionally or alternatively be employed in association with accumulating symbols and activating reel sets in association with one or more plays of a primary game, such as a wagering game. Moreover, while certain embodiments described below are directed to activating reel sets in order from a lowest ranking reel set to highest ranking reel set, it should be appreciated that such embodiments may

alternatively be employed to activate reel sets in order from a highest ranking reel set to lowest ranking reel set. Furthermore, while certain embodiments described below are directed to accumulating symbols and activating reel sets in association with a plurality of reels, it should be appreciated that such embodiments may additionally or alternatively be employed in association with accumulating different game elements and activating sets of game elements, such as, but not limited to, accumulating playing cards and activating sets of playing cards, accumulating keno numbers and activating sets of keno numbers or accumulating bingo elements and activating sets of bingo elements.

[0022] Moreover, while certain embodiments described below are directed to the gaming system accumulating symbols and activating reel sets that are displayed by an electronic gaming machine (“EGM”) such as a slot machine, a video poker machine, a video lottery terminal, a terminal associated with an electronic table game, a terminal associated with a live table game, a video keno machine, a video bingo machine and/or a sports betting terminal that offers sports betting opportunities, it should be appreciated that such embodiments may additionally or alternatively be employed in association with the gaming system accumulating symbols and activating reel sets that are displayed by a personal gaming device (such as desktop computers, laptop computers, tablet computers or computing devices, personal digital assistants, mobile phones, and other mobile computing devices that offer plays of wagering games (and in certain instances, sports betting opportunities)), any suitable slot machine interface board which is in communication with an EGM, and/or any suitable combination of a server operating with an EGM, a slot machine interface board of an EGM and/or a personal gaming device. Furthermore, while the player’s credit balance, the player’s wager, and any awards are displayed as an amount of monetary credits or currency, in certain of the embodiments described below, one or more of such player’s credit balance, such player’s wager, and any awards provided to such a player may be for non-monetary credits, promotional credits, and/or player tracking points or credits.

[0023] FIG. 1 is a flowchart of an example process or method of operating the gaming system of the present disclosure. In various embodiments, the process is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although the process is described with reference to the flowchart shown in FIG. 1, many other processes of performing the acts associated with this illustrated process may be employed. For example, the order of certain of the illustrated blocks or diamonds may be changed, certain of the illustrated blocks or diamonds may be optional, or certain of the illustrated blocks or diamonds may not be employed.

[0024] In various embodiments, the gaming system maintains a plurality of progressive awards as indicated in block 102. In certain embodiments, the plurality of progressive awards are maintained in a multi-level progressive award configuration (“MLP”) associated with a single EGM. In certain embodiments, the progressive awards are maintained in an MLP configuration associated with a plurality of EGMs. In different embodiments wherein the progressive awards are associated with a plurality of EGMs, the EGMs may be in the same bank of EGMs or in the same gaming establishment (such that the progressive award may be

considered a local area progressive (“LAP”)) or in (or otherwise associated with) two or more different gaming establishments (such that the progressive award may be considered a wide area progressive (“WAP”)).

[0025] In certain embodiments, upon an occurrence of a progressive award increment event, the gaming system increases one or more of the maintained progressive awards. In one such embodiment, the progressive award increment event occurs based on (or as a result of) a wager being placed on a play of a game at an EGM associated with that progressive award. In this embodiment, the increase of the maintained progressive award is funded by a portion of the wager placed (e.g., 2% of the amount of wager placed is allocated to growing the progressive award). In another such embodiment, the progressive award increment event additionally or alternatively occurs based on (or as a result of) a side wager being placed at an EGM associated with that progressive award. In this embodiment, the increase of the maintained progressive award is funded by part or all of the side wager placed (e.g., 50% of the amount of side wager placed is allocated to growing the progressive award). In certain embodiments, each of the progressive awards maintained by the gaming system have the same contribution rate (i.e., the portion of the wager placed allocated to the progressive award) and/or the same progressive award reset value. In certain embodiments, a plurality of the progressive awards maintained by the gaming system have different contribution rates and/or different progressive award reset values. In another such embodiment, the progressive award increment event additionally or alternatively occurs based on (or as a result of) one or more displayed events occurring in association with one or more plays of one or more games. In another such embodiment, the progressive award increment event occurs independent of any displayed events associated with any plays of any games.

[0026] In addition to maintaining and periodically increasing the values of the plurality of progressive awards, upon an occurrence of a symbol accumulation sequence triggering event, the gaming system triggers a symbol accumulation sequence as indicated in block 104.

[0027] In certain embodiments, the symbol accumulation sequence comprises a plurality of initial plays of a secondary game, such as a free spin game, wherein the sequence triggering event occurs based on a displayed event associated with a play of a primary game. In another embodiment wherein the symbol accumulation sequence comprises a plurality of initial plays of a secondary game, such as a free spin game, the sequence triggering event occurs based on an event independent of any displayed event associated with the play of the primary game. In certain embodiments, the symbol accumulation sequence comprises a plurality of initial plays of a primary game, such as a primary wagering game, wherein the sequence triggering event includes the placement of a wager on one or more of the plays of the primary game. In certain embodiments, the quantity of initial plays of the game of the triggered symbol accumulation sequence is based on one or more displayed events or elements, such as based on a quantity of triggering symbols that caused the triggering of the symbol accumulation sequence. In certain embodiments, the quantity of initial plays of the game of the triggered symbol accumulation sequence is independent of any displayed events or elements.

[0028] For the triggered symbol accumulation sequence, as indicated in block 106 of FIG. 1, the gaming system displays a plurality of reel sets, wherein each reel set includes a plurality of reels and each reel set is associated with one or more symbol display positions. For example, as seen in FIG. 2A, the gaming system displays five reel sets 202a to 202e, wherein: (i) the first reel set 202a is associated with a plurality of symbol display positions 204a to 204o; (ii) the second reel set 202b is associated with a plurality of symbol display positions 206a to 206d; (iii) the third reel set 202c is associated with a plurality of symbol display positions 208a to 208d; (iv) the fourth reel set 202d is associated with a plurality of symbol display positions 210a to 210d, and (v) the fifth reel set 202e is associated with a plurality of symbol display positions 212a to 212d. In certain embodiments, each reel set is associated with the same quantity of symbol display positions. In certain embodiments, different reel sets are associated with different quantities of symbol display positions.

[0029] Additionally, as indicated in block 106, the gaming system initially designates each reel set as either an active reel set or an inactive reel set. In certain embodiments, for the triggered symbol accumulation sequence, the gaming system determines that one or more reel sets are initially in an active state and zero, one or more reel sets are initially in an inactive state. For example, as seen in FIG. 2A, the gaming system activates a first reel set 202a (i.e., the lowest rank reel set) without activating any of the remaining reel sets 202b to 202e.

[0030] In addition to initially designating each reel set as either an active reel set or an inactive reel set, as also indicated in block 106 of FIG. 1, the gaming system associates each initially inactive reel set with a quantity of points that must be accumulated or earned in association with any active reel sets to activate that initially inactive reel set. In these embodiments, the gaming system associates each initially inactive reel set with a quantity of points required to be accumulated (based on certain symbols displayed at symbol display positions associated with any active reel sets) to cause a modification of that reel set from an inactive state to an active state. For example, as seen in the different threshold levels of the point meter of FIG. 2A, the gaming system associates: (i) initially inactive reel set 202b with 150 points 214a required to be accumulated based on symbols displayed at symbol display positions associated with any active reel sets to activate initially inactive reel set 202b; (ii) initially inactive reel set 202c with 300 points 214b required to be accumulated based on symbols displayed at symbol display positions associated with any active reel sets to activate initially inactive reel set 202c; (iii) initially inactive reel set 202d with 600 points 214c required to be accumulated based on symbols displayed at symbol display positions associated with any active reel sets to activate initially inactive reel set 202d, and (iv) initially inactive reel set 202e with 1000 points 214d required to be accumulated based on symbols displayed at symbol display positions associated with any active reel sets to activate initially inactive reel set 202e.

[0031] In certain embodiments, as further indicated in block 106 of FIG. 1, the gaming system associates each of two or more of the reel sets with one of the maintained progressive awards. In certain embodiments, two or more of the plurality of reels sets are ranked, in order, in accordance with the levels of the progressive awards of the MLP

configuration. For example, as seen in FIG. 2A, of the five displayed reel sets **202a** to **202e**, displayed reel sets **202b** to **202e** are each associated with a different progressive award **216a** to **216d** of the MLP configuration. As seen in this example, the lowest ranked initially inactive reel set **202b** is associated with the lowest level progressive award **216a** of the MLP configuration and the highest ranked initially inactive reel set **202e** is associated with the highest level progressive award **216e** of the MLP configuration.

[0032] For the triggered symbol accumulation sequence, the gaming system determines if the symbol accumulation sequence includes at least one remaining unplayed game as indicated in diamond **108** of FIG. 1. In certain embodiments, the symbol accumulation sequence includes a plurality of initial plays of a game (such as three initial spins) wherein that plurality of initial plays of the game may be supplemented with additional plays of the game, such as responsive to a retriggering event (e.g., an accumulation of a symbol at an available symbol display position). In these embodiments, the gaming system determines if at least one unplayed game remains from the initial plurality of plays of the game of the triggered symbol accumulation sequence or any additional plays of the game provided in association with the symbol accumulation sequence subsequent to the occurrence of the symbol accumulation sequence triggering event. In certain other embodiments, the symbol accumulation sequence includes a plurality of initial plays of a game wherein that plurality of initial plays of the game may not be supplemented with additional plays of the game.

[0033] If the gaming system determines that at least one unplayed game remains associated with the symbol accumulation sequence, at each of zero, one or more symbol display positions not associated with an accumulated symbol, the gaming system determines and displays a symbol as indicated in block **110** of FIG. 1. In these embodiments, the gaming system determines and displays zero, one or more symbols at zero, one or more available symbol display positions (i.e., a symbol display position not associated with a prior accumulated symbol) regardless of the active/inactive status of the reel set associated with such symbol display positions. That is, independent of any determination of whether a reel set (and the individual symbol display positions associated with that reel set) qualifies as an active reel set or an inactive reel set, the gaming system operates to determine and display certain symbols at certain symbol display positions associated with certain reel sets. In certain other embodiments, the gaming system determines and displays zero, one or more symbols at zero, one or more available symbol display positions (i.e., a symbol display position not associated with a prior accumulated symbol) of the active reel sets (without displaying any symbols at any available symbol display positions of the inactive reel sets). In certain embodiments wherein the gaming system employs independent or unsymbol reels (i.e., reels operable to displays a single symbol at a single symbol display position), the gaming system randomly determines, for each available symbol display position, whether or not to display any of a plurality of symbols at that available symbol display position.

[0034] In addition to displaying zero, one or more symbols at zero, one or more available symbol display positions (i.e., symbol display positions not associated with any previously accumulated symbols), as indicated in block **112**, the gaming system accumulates each displayed symbol in association

with the symbol display position where that symbol is displayed at. In certain embodiments, the gaming system operates by accumulating up to one symbol at each symbol display position such that if a symbol was previously accumulated in association with a symbol display position, that accumulated symbol persists for subsequent plays of the game and the gaming system does not display nor accumulate any additional symbols at that symbol display position. In certain other embodiments, the gaming system operates by accumulating up to a designated quantity of at least two symbols at each symbol display position such that if the designated quantity of symbols were previously accumulated in association with a symbol display position, those accumulated symbols persist for subsequent plays of the game and the gaming system does not display nor accumulate any additional symbols at that symbol display position.

[0035] In certain embodiments, each symbol is associated with an award. In these embodiments, an award associated with a determined and displayed symbol may contribute to a total award for the triggered symbol accumulation sequence based on whether or not the reel set associated with the symbol display position where that symbol is accumulated at is an active reel set or an inactive reel set. In operation of certain embodiments, the gaming system randomly determines which symbols, if any, to display at which symbol display positions based on the award of that symbol. For example, a symbol with a relatively low award has a first probability of being randomly displayed at a symbol display position, and a symbol with a relatively high award has a second, lower probability of being randomly displayed at a symbol display position. In operation of certain embodiments, the gaming system additionally or alternatively randomly determines which symbols, if any, to display at which symbol display positions based on the symbol display position. For example, a symbol with a relatively high award has a first probability of being randomly displayed at a symbol display position associated with a relatively lower ranked reel set and a second, greater probability of being randomly displayed at a symbol display position associated with a relatively higher ranked reel set.

[0036] In certain embodiments, certain symbols, such as designated symbols, are associated with a quantity of points. In these embodiments, a quantity of points associated with a determined and displayed designated symbol may contribute to a total quantity of points (for activating addition reel sets) based on whether or not the reel set associated with the symbol display position where that designated symbol is accumulated at is an active reel set or an inactive reel set. In operation of certain embodiments, the gaming system randomly determines which designated symbols, if any, to display at which symbol display positions based on the quantity of points associated with that symbol. For example, a designated symbol with a relatively low quantity of points has a first probability of being randomly displayed at a symbol display position, and a designated symbol with a relatively high quantity of points has a second, lower probability of being randomly displayed at a symbol display position. In operation of certain embodiments, the gaming system additionally or alternatively randomly determines which designated symbols, if any, to display at which symbol display positions based on the symbol display position. For example, a designated symbol with a relatively high quantity of points has a first probability of being randomly displayed at a symbol display position associated

with a relatively lower ranked reel set and a second, greater probability of being randomly displayed at a symbol display position associated with a relatively higher ranked reel set.

[0037] Returning to FIG. 2A, following the triggering of a symbol accumulation sequence initially including three plays of the game and the determination that at least one play of the game remains of the triggered symbol accumulation sequence, for a first play of a game of the symbol accumulation sequence, the gaming system spins: (i) a plurality of unisymbol reels of the first, initially active reel set **202a** that are associated with the plurality of symbol display positions **204a** to **204o**; (ii) a plurality of unisymbol reels of the second, initially inactive reel set **202b** that are associated with the plurality of symbol display positions **206a** to **206d**; (iii) a plurality of unisymbol reels of the third, initially inactive reel set **202c** that are associated with the plurality of symbol display positions **208a** to **208d**; (iv) a plurality of unisymbol reels of the fourth, initially inactive reel set **202d** that are associated with the plurality of symbol display positions **210a** to **210d**; and (v) a plurality of unisymbol reels of the fifth, initially inactive reel set **202e** that are associated with the plurality of symbol display positions **212a** to **212d**. As seen in FIG. 2B of this example, when stopped, each unisymbol reel may or may not display a symbol **218** associated with an award and/or a quantity of points. For this first play of the game of the symbol accumulation sequence wherein each of the symbol display positions are available to potentially display a symbol (i.e., none of the symbol display positions are unavailable due to being associated with a previously accumulated symbol), the gaming system displays symbols **218a** to **218d**, wherein each of the symbols are associated with awards of varying values and symbol **218a** is additionally associated with a quantity of 50 points. The gaming system accumulates each of these four symbols in association with its respective symbol display position. In this example, since at least one symbol was accumulated from this play of the game, the gaming system resets the quantity of plays of the game remaining for the triggered symbol accumulation sequence back to three plays of the game.

[0038] In certain embodiments, the gaming system displays an associated quantity of points when displaying (or alternatively after displaying) a symbol at a symbol display position. In certain embodiments, in association with the display of a designated symbol associated with a quantity of points, the gaming system displays a play of a game, such as a random point count-up, that results in the display of a quantity of points associated with a symbol displayed at a symbol display position.

[0039] In this example, while only the first reel set **202a** is active for this first play of the game of the triggered symbol accumulation sequence, the gaming system determines, displays and accumulates symbols (i.e., symbols **218a** and **218b**) at available symbol display positions associated with active reel set **202a** and symbols (i.e., symbols **218c** to **218d**) at available symbol display positions associated with the remaining inactive reel sets **202b** to **202e**. As such, in this example, the activation or inactivation of any particular reel set does not factor into whether or not any symbols may be determined, displayed and accumulated at the available symbol display positions associated with that particular reel set. Such a configuration provides that the gaming system determines and displays certain symbols associated with certain awards and/or points at certain symbol display

positions which may or may not be part of subsequently activated reel sets to increase the level of engagement that the user has with the gaming system.

[0040] In addition to accumulating zero, one or more symbols, as indicated in block **114** of FIG. 1, the gaming system causes each symbol that is associated with a quantity of points and is accumulated at a symbol display position associated with an active reel set to contribute to the quantity of points that must be accumulated to activate any inactive reel sets. In these embodiments, the gaming system employs the points associated with the symbols accumulated at the symbol display positions of the active set(s) of reels to potentially activate additional otherwise inactive reel sets. That is, the gaming system accounts for the accumulation of symbols associated with points at symbol display positions associated with any active reel sets in determining whether or not the quantity of accumulated symbols has been achieved for any inactive reel sets. For example, as seen in FIG. 2B, to account for designated symbol **218a** at symbol display position **204d** associated with the active first reel set **202a** being associated with 50 points, the gaming system displays an increase of the point meter by 50 points and reduces the quantity of points required to activate each of the inactive reel sets **202b** to **202e** by 50 points. As such, following the accumulation of designated symbol **218a** at a symbol display position associated with the active reel set as part of the first play of the game of the symbol accumulation sequence, initially inactive reel set **202b** now requires 100 more points to be activated; initially inactive reel set **202c** now requires 250 more points to be activated; initially inactive reel set **202d** now requires 550 more points to be activated; and initially inactive reel set **202e** now requires 950 more points to be activated.

[0041] It should be appreciated that in these embodiments, while symbols may be accumulated at symbol display positions associated with active reel sets or inactive reel sets, the gaming system utilizes points associated with symbols accumulated at symbol display positions associated with active reel sets when tracking quantities of points to potentially activate otherwise inactive reel sets. In such embodiments, any points associated with symbols displayed at symbol display positions associated with inactive reel sets are not utilized to reduce any quantities of points needed to activate any inactive reel sets. In certain other embodiments, the gaming system utilizes points associated with symbols accumulated at symbol display positions associated with active reel sets and inactive reel sets when tracking quantities of points to potentially activate otherwise inactive reel sets.

[0042] Following the accumulation of symbols at zero, one or more symbol display positions associated with zero, one or more reel sets (regardless of the state of each reel set being in an active state or an inactive state) and the accounting of any points associated with any symbols accumulated at any symbol display positions associated with any active reel sets toward the goal quantity of points need to activate zero, one or more of any inactive reel sets, the gaming system determines if a reel set activation event has occurred in association with any of the inactive reel sets as indicated in diamond **116** of FIG. 1. In certain embodiments, a reel set activation event occurs in association with an inactive reel set if the quantity of points required to be accumulated (based on symbols displayed at symbol display positions associated with any active reel sets) associated with that

inactive reel set has been achieved. For example, referring back to FIG. 2A, a reel set activation occurs: (i) for inactive reel set **202b** if 150 points are earned; (ii) for inactive reel set **202c** if 300 points are earned; (iii) for inactive reel set **202d** if 600 points are earned; and (iv) for inactive reel set **202e** if 1000 points are earned.

[0043] If the gaming system determines that no reel set activation event occurred in association with any inactive reel sets, the gaming system returns to diamond **108** and again determines if the symbol accumulation sequence includes at least one remaining unplayed game. For example, as seen in FIG. 2B, upon the gaming system determining that none of the required quantities of points associated with any inactive reel sets have been accumulated based on symbols displayed at symbol display positions associated with active reel sets (i.e., less than 150 points are earned to activate inactive reel set **202b**; less than 300 points are earned to activate inactive reel set **202c**; less than 600 points are earned to activate inactive reel set **202d**; and less than 1000 points are earned to activate inactive reel set **202e**), the gaming system does not cause any reel set activation event to occur. In this case, the gaming system proceeds to determine if at least one unplayed game of the triggered symbol accumulation sequence remains and if so, displays another play of the game of the triggered symbol accumulation sequence.

[0044] For the next play of the game of the triggered symbol accumulation sequence, the gaming system again determines, displays and accumulates zero, one or more symbols at zero, one or more available symbol display positions as described above. For example, as seen in FIGS. 2C and 2D, for another play of the game of the triggered symbol accumulation sequence, the gaming system spins the plurality of unisymbol reels associated with the plurality of symbol display positions **204**, **206**, **208**, **210** and **212** not associated with any accumulated symbols (FIG. 2C). For this second play of the game of the symbol accumulation sequence only certain of the symbol display positions are available to potentially display a symbol (i.e., certain of the symbol display positions are unavailable due to being associated with a previously accumulated symbol). In this example, when these reels stop spinning, the gaming system displays symbols **218e** to **218k** associated with awards and/or points (FIG. 2D). The gaming system accumulates each of these additional symbols in association with its respective symbol display position. In this example, since at least one symbol was accumulated from this play of the game, the gaming system resets the quantity of plays of the game remaining for the triggered symbol accumulation sequence back to three plays of the game.

[0045] In addition to accumulating such displayed symbols, the gaming system again determines if a reel set activation event has occurred in association with any of the inactive reel sets. As described above, if the gaming system determines that no reel set activation event occurred in association with any inactive reel sets, the gaming system returns to diamond **108** and again determines if the symbol accumulation sequence includes at least one remaining unplayed game. On the other hand, if the gaming system determines that a reel set activation event occurred in association with an inactive reel set, the gaming system activates the reel set in which a reel set activation event occurred in association with as indicated in block **118** of FIG. 1. In certain embodiments wherein a reel set activation

event occurs in association with an inactive reel set if the quantity of points required to be accumulated (based on symbols displayed at symbol display positions associated with any active reel sets) associated with that inactive reel set has been achieved, if the associated quantity of points has been accumulated, the gaming system activates that reel set. For example, as seen in FIG. 2D, since 150 points have been earned based on symbols accumulated at symbol display positions associated with the first active reel set **202a**, the gaming system determines that a reel set activation event has occurred in association with the second inactive reel sets and the gaming system activates reel set **202b**.

[0046] Following this activation of one or more reel sets, the gaming system returns to diamond **116** and again determines if a reel set activation event has occurred in association with any inactive reel sets. That is, since the activation of a previously inactive reel set may result in a modification of the quantity of points associated with symbols displayed at symbol display positions associated with active reel sets, the activation of one reel set may cause a chain reaction wherein one or more additional reel sets are also activated. Put differently, while any points associated with symbols displayed at symbol display positions associated with a then inactive reel set were not previously accounted for in determining whether or not a reel set activation event occurred, any points associated with any accumulated symbols displayed symbols at any symbol display positions associated with the now active reel set are accounted for in determining whether or not a reel set activation event occurred with any inactive reel sets. In other words, while the determination, display and accumulation of points associated with a symbol displayed at a symbol display position associated with an inactive reel set does not contribute to the determination of if a quantity of points causes the activation of any inactive reel set when such points are accumulated, the points associated with that symbol may subsequently contribute to the determination of if a quantity of points causes the activation of any inactive reel set at a later point in time during the triggered sequence. This configuration provides that points associated with a symbol accumulated at a symbol display position associated with an inactive reel set may operate to at least partially provide a delayed activation of an inactive reel set, wherein such a delayed activation depends on one or more events otherwise occurring during the triggered sequence. For example, as seen in FIG. 2E, since the accumulation of 150 points associated with symbols displayed at symbol display positions associated with active reel set **202a** resulted in the activation of reel set **202b**, and since symbol **218h** associated with 100 points was previously accumulated at symbol display position **206d** associated with reel set **202b**, the activation of reel set **202b** resulted in the accumulation of another 100 points toward activating the next reel set. As seen in this example, to account for designated symbol **218h** at symbol display position **206d** associated with the now active second reel set **202b** being associated with 100 points, the gaming system displays an increase of the point meter by 100 points and reduces the quantity of points required to activate each of the remaining inactive reel sets **202c** to **202e** by 100 points.

[0047] Following any subsequent activation of any inactive reel set or if the gaming system determines that no reel set activation event occurred in association with any inactive reel sets, the gaming system returns to diamond **108** and

again determines if the symbol accumulation sequence includes at least one remaining unplayed game.

[0048] Upon the gaming system determining that no games remain in the triggered symbol accumulation sequence, as indicated in blocks 120 and 122 of FIG. 1, the gaming system determines and displays a total award based on any awards of any accumulated symbols at any symbol display positions of any active reel sets and further based on any progressive awards associated with any active reel sets in which a progressive award triggering event occurred. That is, upon an occurrence of a triggered symbol accumulation sequence award determination event, the gaming system proceeds to determine a total award for the triggered symbol accumulation sequence wherein the active or inactive state of each reel set determines whether or not any awards may be available in association with that reel set.

[0049] In certain embodiments, the total award is based on any awards of any accumulated symbols at any symbol display positions of any active reel sets. For example, as seen in FIG. 2F, following the occurrence of one or more progressive award increment events that each increased the values of the maintained progressive awards 216a to 216d, and following additional plays of the game of the triggered symbol accumulation sequence that resulted in the determination, display and accumulation of additional symbols 2181 to 2180, when no more games remain to be played in the triggered symbol accumulation sequence, the gaming system determines a total award based on the awards associated with the symbols accumulated at the symbol display positions of the active reel sets. In this example, since reel sets 202a to 202c are each active, the gaming system determines a total award of six-thousand-nine-hundred credits based on the awards of the symbols accumulated at symbol display positions of reel sets 202a to 202c. It should be appreciated that in these embodiments, in determining the total award, the gaming system does not factor in any awards of any accumulated symbols at any symbol display positions of any inactive reel sets. For example, since reel sets 202d and 202e are each inactive, the gaming system does not account for the awards of the symbols accumulated at symbol display positions of reel sets 202d and 202e in determining the total award. Such a configuration provides that the graphical user interface employed by the gaming system displays certain awards associated with certain symbols even though such awards are not ultimately provided, to increase the amount of anticipation certain users experience.

[0050] In certain embodiments, in addition to or alternative from the total award being based on any awards of any accumulated symbols at any symbol display positions of any active reel sets, the total award is based on any progressive awards associated with any active reel sets in which at least a designated quantity of symbols are accumulated at a designated quantity of symbol display positions of that reel set. In these embodiments, the gaming system determines whether or not to include zero, one or more progressive awards in the total award based on whether or not a requisite quantity of symbols have been accumulated at a requisite quantity of symbol display positions of zero, one or more active reel sets. In other words, the gaming system determines whether or not a progressive award triggering event occurs in association with zero, one or more active reel sets based on the accumulation of symbols at the symbol display positions of such active reel sets. For example, as also seen

in FIG. 2F, the gaming system additionally determines the total award based on a progressive award triggering event occurring in association with progressive award 216a. In this example, since symbols are accumulated at each of the symbol display positions 206a to 206d of active reel set 202a, the gaming system determines to include in the total award progressive award 216a (currently valued at \$12.14) associated with this complete reel set. In this example, following the gaming system providing any progressive awards in association with the triggered symbol accumulation sequence, the gaming system resets the value of each such progressive award to a respective reset value (not shown).

[0051] In certain embodiments, for a progressive award associated with an active reel set to be included in the total award, the gaming system requires that a symbol be accumulated at each of the symbol display positions of that active reel set. In certain other embodiments, for a progressive award associated with an active reel set to be included in the total award, the gaming system requires that a symbol be accumulated at less than each of the symbol display positions of that active reel set.

[0052] It should be appreciated that in these embodiments, in determining the total award, the gaming system does not factor in any progressive awards associated with any inactive reel sets regardless of any accumulated symbols at any symbol display positions of such inactive reel sets. That is, the gaming system does not account for the progressive award associated with an inactive reel set even though a symbol has been accumulated at each of the symbol display positions of this inactive reel set. Such a configuration provides that the graphical user interface employed by the gaming system displays certain progressive award that would be provided only if one or more symbols independent of the progressive award were accumulated during the triggered sequence.

[0053] In certain embodiments, zero, one or more awards associated with one or more symbols and/or a total award determined in association with a triggered symbol accumulation sequence includes one or more of: a quantity of monetary credits, a quantity of non-monetary credits, a quantity of promotional credits, a quantity of player tracking points, a progressive award, a modifier, such as a multiplier, a quantity of free plays of one or more games, a quantity of plays of one or more secondary or bonus games, a multiplier of a quantity of free plays of a game, one or more lottery based awards, such as lottery or drawing tickets, a wager match for one or more plays of one or more games, an increase in the average expected payback percentage for one or more plays of one or more games, one or more comps (e.g., a free dinner, a free night's stay at a hotel), a relatively high value product (e.g., a free car), one or more bonus credits usable for online play, a lump sum of player tracking points or credits, a multiplier for player tracking points or credits, an increase in a membership or player tracking level, one or more coupons or promotions usable within and/or outside of the gaming establishment (e.g., a 20% off coupon for use at a convenience store), virtual goods associated with the gaming system, virtual goods not associated with the gaming system, an access code usable to unlock content on an internet.

[0054] In certain embodiments, zero, one or more awards associated with one or more symbols and/or a total award determined in association with a triggered symbol accumu-

lation sequence includes an activation of a feature of a game. In different embodiments, such features include any feature that results in a modification of one or more components, aspects, or elements of one or more subsequent plays of a game, such as the modification of one or more game outcomes of one or more plays of a game (e.g., the symbols evaluated for the play(s) of the game), the modification of the payable utilized for one or more plays of the game and/or the modification of any award determined for one or more plays of the game. In different embodiments, such features include, but are not limited to: a feature which superimposed one or more symbols over the randomly generated symbols of the reels; a feature which replaces one or more symbols of the randomly generated symbols of the reels with a predetermined symbol pattern; a feature which replaces one or more symbols of the randomly generated symbols of the reels with a predetermined pattern of wild symbols; a modifier, such as a multiplier, feature; a book-end wild symbols feature; a stacked wild symbols feature; an expanding wild symbols feature; a nudging wild symbols feature; a feature modifying a quantity of wild symbols available to be generated; a retrigger symbol feature; an anti-terminator symbol feature; a locking reel feature; an expanding reel feature; a locking symbol position feature; a feature modifying a placed wager amount; a feature modifying a placed side wager amount; a feature modifying a number of wagered on paylines; a feature modifying a wager placed on one or more paylines (or on one or more designated paylines); a feature modifying a number of ways to win wagered on; a feature modifying a wager placed on one or more ways to win (or on one or more designated ways to win); a feature modifying a payable utilized for a play of a game; a feature modifying an average expected payback percentage of a play of a game; a feature modifying an average expected payout of a play of a game; a feature modifying one or more awards available; a feature modifying a range of awards available; a feature modifying a type of awards available; a feature modifying one or more progressive awards; a feature modifying which progressive awards are available to be won; a feature modifying one or more modifiers, such as multipliers, available; a feature modifying an activation of a reel (or a designated reel); a feature modifying an activation of a plurality of reels; a feature modifying a generated outcome (or a designated generated outcome); a feature modifying a generated outcome (or a designated generated outcome) associated with an award over a designated value; a feature modifying a generated outcome (or a designated generated outcome) on a designated payable; a feature modifying a generated outcome (or a designated generated outcome) in a scatter configuration; a feature modifying a winning way to win (or a designated winning way to win); a feature modifying a designated symbol or symbol combination; a feature modifying a generation of a designated symbol or symbol combination on a designated payable; a feature modifying a generation of a designated symbol or symbol combination in a scatter configuration; a feature modifying a quantity of picks in a selection game; a feature modifying a quantity of offers in an offer and acceptance game; a feature modifying a quantity of moves in a trail game; a feature modifying an amount of free spins provided; a feature modifying a game terminating or ending condition; a feature modifying how one or more aspects of one or more games (e.g., colors, speeds, sound) are displayed to a player; and/or a feature

modifying any game play feature associated with any play of any game of the present disclosure.

[0055] In certain embodiments, the gaming system initially displays the initially active reel sets and the initially inactive reel sets. In certain embodiments, the gaming system initially displays the initially active reel sets while not displaying any initially inactive reel sets. In these embodiments, as zero, one or more additional reel sets become activated, the gaming system displays such activated reels sets spawning from previously active reel sets.

[0056] In certain embodiments, the gaming system initially displays the initially active reel sets and the initially inactive reel sets as the same size. In certain embodiments, the gaming system initially displays the initially active reel sets and the initially inactive reel sets as different sizes. In these embodiments, as zero, one or more additional reel sets become activated, the gaming system modifies the size of such activated reel sets to draw the user's attention to the activated reel set.

[0057] In one embodiment, the gaming system causes at least one display device of the gaming system to display the triggered symbol accumulation sequence. In another embodiment, the gaming system additionally or alternatively causes one or more community or overhead display devices to display part or all of the triggered symbol accumulation sequence to one or more other players or bystanders either at a gaming establishment or viewing over a network, such as the internet. In another embodiment, the gaming system additionally or alternatively causes one or more internet sites to each display the triggered symbol accumulation sequence such that a player is enabled to log on from a personal web browser.

[0058] In different embodiments, as described above, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on an outcome associated with one or more plays of any primary games or any secondary games. In one embodiment, such determinations are symbol driven based on the generation of one or more designated symbols or symbol combinations. In various embodiments, a generation of a designated symbol (or sub-symbol) or a designated set of symbols (or sub-symbols) over one or more plays of a primary game (and/or a secondary game) causes such conditions to be satisfied and/or one or more of such events to occur.

[0059] In different embodiments, the gaming system does not provide any apparent reasons to the players for an occurrence of a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set. In these embodiments, such determinations are not triggered by an event in a primary game or based specifically on any of the plays of any primary games or any secondary games. That is, these events occur without any explanation or alternatively with simple explanations.

[0060] In one such embodiment, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on an amount of coin-in. In this embodiment, the gaming system determines if an amount of coin-in reaches or exceeds a designated amount of coin-in

(i.e., a threshold coin-in amount). Upon the amount of coin-in wagered reaching or exceeding the threshold coin-in amount, the gaming system causes one or more of such events or conditions to occur. In another such embodiment, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on an amount of virtual currency-in. In this embodiment, the gaming system determines if an amount of virtual currency-in wagered reaches or exceeds a designated amount of virtual currency-in (i.e., a threshold virtual currency-in amount). Upon the amount of virtual currency-in wagered reaching or exceeding the threshold virtual currency-in amount, the gaming system causes one or more of such events or conditions to occur. In different embodiments, the threshold coin-in amount and/or the threshold virtual currency-in amount is predetermined, randomly determined, determined based on a player's status (such as determined through a player tracking system), determined based on a generated symbol or symbol combination, determined based on a random determination by the central controller, determined based on a random determination at the gaming device, determined based on one or more side wagers placed, determined based on the player's primary game wager, determined based on time (such as the time of day) or determined based on any other suitable method or criteria.

[0061] In one such embodiment, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on an amount of coin-out. In this embodiment, the gaming system determines if an amount of coin-out reaches or exceeds a designated amount of coin-out (i.e., a threshold coin-out amount). Upon the amount of coin-out reaching or exceeding the threshold coin-out amount, the gaming system causes one or more of such events or conditions to occur. In another such embodiment, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on an amount of virtual currency-out. In this embodiment, the gaming system determines if an amount of virtual currency-out reaches or exceeds a designated amount of virtual currency-out (i.e., a threshold virtual currency-out amount). Upon the amount of virtual currency-out reaching or exceeding the threshold virtual currency-out amount, the gaming system causes one or more of such events or conditions to occur. In different embodiments, the threshold coin-out amount and/or the threshold virtual currency-out amount is predetermined, randomly determined, determined based on a player's status (such as determined through a player tracking system), determined based on a generated symbol or symbol combination, determined based on a random determination by the central controller, determined based on a random determination at the gaming device, determined based on one or more side wagers placed, determined based on the player's primary game wager, determined based on time (such as the time of day) or determined based on any other suitable method or criteria.

[0062] In different embodiments, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award

triggering event for a progressive award associated with an active reel set occurs based on a predefined variable reaching a defined parameter threshold. For example, when the 500,000th player has played an EGM (ascertained from a player tracking system), one or more of such events or conditions occur. In different embodiments, the predefined parameter thresholds include a length of time, a length of time after a certain dollar amount is hit, a wager level threshold for a specific device (which gaming device is the first to contribute \$250,000), a number of gaming devices active, or any other parameter that defines a suitable threshold.

[0063] In different embodiments, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on a quantity of games played. In this embodiment, a quantity of games played is set for when one or more of such events or conditions will occur. In one embodiment, such a set quantity of games played is based on historic data.

[0064] In different embodiments, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on time. In this embodiment, a time is set for when one or more of such events or conditions will occur. In one embodiment, such a set time is based on historic data.

[0065] In different embodiments, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based upon gaming system operator defined player eligibility parameters stored on a player tracking system (such as via a player tracking card or other suitable manner). In this embodiment, the parameters for eligibility are defined by the gaming system operator based on any suitable criterion. In one embodiment, the gaming system recognizes the player's identification (via the player tracking system) when the player inserts or otherwise associates their player tracking card in the EGM and/or logs into the player tracking system using a mobile device, such as a personal gaming device. The gaming system determines the player tracking level of the player and if the current player tracking level defined by the gaming system operator is eligible for one or more of such events or conditions. In one embodiment, the gaming system operator defines minimum bet levels required for such events or conditions to occur based on the player's card level.

[0066] In different embodiments, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on a system determination, including one or more random selections by the central controller. For example, as described above, the gaming system tracks all active EGMs and the wagers they placed, wherein based on the EGM's state as well as one or more wager pools associated with the EGM, the gaming system determines whether to one or more of such events or conditions will occur. In one such embodiment, the player who consistently places a higher wager is more likely to be associated with an occurrence of one or more of such events

or conditions than a player who consistently places a minimum wager. It should be appreciated that the criteria for determining whether a player is in active status or inactive status for determining if one or more of such events occur may be the same as, substantially the same as, or different than the criteria for determining whether a player is in active status or inactive status for another one of such events to occur.

[0067] In different embodiments, a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs based on a determination of if any numbers allotted to a gaming device match a randomly selected number. In this embodiment, upon or prior to each play of each gaming device, a gaming device selects a random number from a range of numbers and during each primary game, the gaming device allocates the first N numbers in the range, where N is the number of credits bet by the player in that primary game. At the end of the primary game, the randomly selected number is compared with the numbers allocated to the player and if a match occurs, one or more of such events or conditions occur.

[0068] It should be appreciated that any suitable manner of causing a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set to occur may be implemented in accordance with the gaming system and method of the present disclosure. It should be further appreciated that one or more of the above-described triggers pertaining to a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurring may be combined in one or more different embodiments.

[0069] It should be appreciated that in different embodiments, one or more of: when a symbol accumulation sequence triggering event, a progressive award increment event, a reel set activation event and/or a progressive award triggering event for a progressive award associated with an active reel set occurs; a quantity of reel sets to employ during the triggered symbol accumulation sequence; a quantity of symbol display positions to associate with each reel set; which reel sets are initially active and initially inactive upon the triggering of the symbol accumulation sequence; which progressive awards are associated with which reel sets; a quantity of points associated with each inactive reel set that, if accumulated based on points associated with symbols displayed at symbol display positions associated with active reel sets, cause a reel set activation event to occur for that inactive reel set; which symbols to display at which symbol display positions; which awards to associate with which symbols; a quantity of symbols to accumulate at a symbol display position; and/or any determination of the present disclosure is/are predetermined, randomly determined, randomly determined based on one or more weighted percentages, determined based on a symbol or symbol combination, determined independent of any symbols or symbol combinations, determined based on a random determination by a server, determined independent of a random determination by a server, determined based on at least one play of at least one game, determined independent of at least

one play of at least one game, determined based on a player's selection, determined independent of a player's selection, determined based on one or more side wagers placed, determined independent of one or more side wagers placed, determined based on the player's primary game wager, determined independent of the player's primary game wager, determined based on time (such as the time of day), determined independent of time (such as the time of day), determined based on an amount of coin-in accumulated in one or more pools, determined independent of an amount of coin-in accumulated in one or more pools, determined based on a status of the player (i.e., a player tracking status), determined independent of a status of the player (i.e., a player tracking status), determined based on one or more other determinations of the present disclosure, determined independent of any other determination of the present disclosure or determined based on any other suitable method or criteria.

[0070] The above-described embodiments of the present disclosure may be implemented in accordance with or in conjunction with one or more of a variety of different types of gaming systems, such as, but not limited to, those described below.

[0071] The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. A "gaming system" as used herein refers to various configurations of: (a) one or more servers; (b) one or more electronic gaming machines such as those located on a casino floor; and/or (c) one or more personal gaming devices. Thus, in various embodiments, the gaming system of the present disclosure includes: (a) one or more electronic gaming machines in combination with one or more servers; (b) one or more personal gaming devices in combination with one or more servers; (c) one or more personal gaming devices in combination with one or more electronic gaming machines; (d) one or more personal gaming devices, one or more electronic gaming machines, and one or more servers in combination with one another; (e) a single electronic gaming machine; (f) a plurality of electronic gaming machines in combination with one another; (g) a single personal gaming device; (h) a plurality of personal gaming devices in combination with one another; (i) a single server; and/or (j) a plurality of servers in combination with one another. For brevity and clarity and unless specifically stated otherwise, "EGM" as used herein represents one EGM or a plurality of EGMs, "personal gaming device" as used herein represents one personal gaming device or a plurality of personal gaming devices, and "server" as used herein represents one server or a plurality of servers.

[0072] As noted above, in various embodiments, the gaming system includes an EGM (or personal gaming device) in combination with a server. In such embodiments, the EGM (or personal gaming device) is configured to communicate with the server through a data network or remote communication link. In certain such embodiments, the EGM (or personal gaming device) is configured to communicate with another EGM (or personal gaming device) through the same data network or remote communication link or through a different data network or remote communication link. For example, the gaming system includes a plurality of EGMs that are each configured to communicate with a server through a data network.

[0073] In certain embodiments in which the gaming system includes an EGM (or personal gaming device) in combination with a server, the server is any suitable computing device (such as a server) that includes at least one processor and at least one memory device or data storage device. As further described herein, the EGM (or personal gaming device) includes at least one EGM (or personal gaming device) processor configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the EGM (or personal gaming device) and the server. The at least one processor of that EGM (or personal gaming device) is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the EGM (or personal gaming device). Moreover, the at least one processor of the server is configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the server and the EGM (or personal gaming device). The at least one processor of the server is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the server. One, more than one, or each of the functions of the server may be performed by the at least one processor of the EGM (or personal gaming device). Further, one, more than one, or each of the functions of the at least one processor of the EGM (or personal gaming device) may be performed by the at least one processor of the server.

[0074] In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the EGM (or personal gaming device) are executed by the server. In such “thin client” embodiments, the server remotely controls any games (or other suitable interfaces) displayed by the EGM (or personal gaming device), and the EGM (or personal gaming device) is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the EGM (or personal gaming device) are communicated from the server to the EGM (or personal gaming device) and are stored in at least one memory device of the EGM (or personal gaming device). In such “thick client” embodiments, the at least one processor of the EGM (or personal gaming device) executes the computerized instructions to control any games (or other suitable interfaces) displayed by the EGM (or personal gaming device).

[0075] In various embodiments in which the gaming system includes a plurality of EGMs (or personal gaming devices), one or more of the EGMs (or personal gaming devices) are thin client EGMs (or personal gaming devices) and one or more of the EGMs (or personal gaming devices) are thick client EGMs (or personal gaming devices). In other embodiments in which the gaming system includes one or more EGMs (or personal gaming devices), certain functions of one or more of the EGMs (or personal gaming devices) are implemented in a thin client environment, and certain other functions of one or more of the EGMs (or personal gaming devices) are implemented in a thick client environment. In one such embodiment in which the gaming system includes an EGM (or personal gaming device) and a server, computerized instructions for controlling any primary or base games displayed by the EGM (or personal gaming device) are communicated from the server to the EGM (or

personal gaming device) in a thick client configuration, and computerized instructions for controlling any secondary or bonus games or other functions displayed by the EGM (or personal gaming device) are executed by the server in a thin client configuration.

[0076] In certain embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a local area network (LAN) in which the EGMs (or personal gaming devices) are located substantially proximate to one another and/or the server. In one example, the EGMs (or personal gaming devices) and the server are located in a gaming establishment or a portion of a gaming establishment.

[0077] In other embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a wide area network (WAN) in which one or more of the EGMs (or personal gaming devices) are not necessarily located substantially proximate to another one of the EGMs (or personal gaming devices) and/or the server. For example, one or more of the EGMs (or personal gaming devices) are located: (a) in an area of a gaming establishment different from an area of the gaming establishment in which the server is located; or (b) in a gaming establishment different from the gaming establishment in which the server is located. In another example, the server is not located within a gaming establishment in which the EGMs (or personal gaming devices) are located. In certain embodiments in which the data network is a WAN, the gaming system includes a server and an EGM (or personal gaming device) each located in a different gaming establishment in a same geographic area, such as a same city or a same state. Gaming systems in which the data network is a WAN are substantially identical to gaming systems in which the data network is a LAN, though the quantity of EGMs (or personal gaming devices) in such gaming systems may vary relative to one another.

[0078] In further embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is an internet (such as the Internet) or an intranet. In certain such embodiments, an Internet browser of the EGM (or personal gaming device) is usable to access an Internet game page from any location where an Internet connection is available. In one such embodiment, after the EGM (or personal gaming device) accesses the Internet game page, the server identifies a player before enabling that player to place any wagers on any plays of any wagering games. In one example, the server identifies the player by requiring a player account of the player to be logged into via an input of a unique username and password combination assigned to the player. The server may, however, identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identi-

fication number associated with the player by the server; or by identifying the EGM (or personal gaming device), such as by identifying the MAC address or the IP address of the Internet facilitator. In various embodiments, once the server identifies the player, the server enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games, and displays those plays via the Internet browser of the EGM (or personal gaming device).

[0079] The server and the EGM (or personal gaming device) are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via: a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile Internet network), or any other suitable medium. The expansion in the quantity of computing devices and the quantity and speed of Internet connections in recent years increases opportunities for players to use a variety of EGMs (or personal gaming devices) to play games from an ever-increasing quantity of remote sites. Additionally, the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

[0080] FIG. 3 is a block diagram of an example EGM 1000 and FIGS. 4A and 4B include two different example EGMs 2000a and 2000b. The EGMs 1000, 2000a, and 2000b are merely example EGMs, and different EGMs may be implemented using different combinations of the components shown in the EGMs 1000, 2000a, and 2000b. Although the below refers to EGMs, in various embodiments personal gaming devices (such as personal gaming device 2000c of FIG. 4C) may include some or all of the below components.

[0081] In these embodiments, the EGM 1000 includes a master gaming controller 1012 configured to communicate with and to operate with a plurality of peripheral devices 1022.

[0082] The master gaming controller 1012 includes at least one processor 1010. The at least one processor 1010 is any suitable processing device or set of processing devices, such as a microprocessor, a microcontroller-based platform, a suitable integrated circuit, or one or more application-specific integrated circuits (ASICs), configured to execute software enabling various configuration and reconfiguration tasks, such as: (1) communicating with a remote source (such as a server that stores authentication information or game information) via a communication interface 1006 of the master gaming controller 1012; (2) converting signals read by an interface to a format corresponding to that used by software or memory of the EGM; (3) accessing memory to configure or reconfigure game parameters in the memory according to indicia read from the EGM; (4) communicating with interfaces and the peripheral devices 1022 (such as input/output devices); and/or (5) controlling the peripheral devices 1022. In certain embodiments, one or more components of the master gaming controller 1012 (such as the at least one processor 1010) reside within a housing of the EGM (described below), while in other embodiments at

least one component of the master gaming controller 1012 resides outside of the housing of the EGM.

[0083] The master gaming controller 1012 also includes at least one memory device 1016, which includes: (1) volatile memory (e.g., RAM 1009, which can include non-volatile RAM, magnetic RAM, ferroelectric RAM, and any other suitable forms); (2) non-volatile memory 1019 (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.); (3) unalterable memory (e.g., EPROMs 1008); (4) read-only memory; and/or (5) a secondary memory storage device 1015, such as a non-volatile memory device, configured to store gaming software related information (the gaming software related information and the memory may be used to store various audio files and games not currently being used and invoked in a configuration or reconfiguration). Any other suitable magnetic, optical, and/or semiconductor memory may operate in conjunction with the EGM of the present disclosure. In certain embodiments, the at least one memory device 1016 resides within the housing of the EGM (described below), while in other embodiments at least one component of the at least one memory device 1016 resides outside of the housing of the EGM. In these embodiments, any combination of one or more computer readable media may be utilized. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0084] A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

[0085] The at least one memory device 1016 is configured to store, for example: (1) configuration software 1014, such as all the parameters and settings for a game playable on the EGM; (2) associations 1018 between configuration indicia read from an EGM with one or more parameters and settings; (3) communication protocols configured to enable

the at least one processor **1010** to communicate with the peripheral devices **1022**; and/or (4) communication transport protocols (such as TCP/IP, USB, Firewire, IEEE1394, Bluetooth, IEEE 802.11x (IEEE 802.11 standards), hiperlan/2, HomeRF, etc.) configured to enable the EGM to communicate with local and non-local devices using such protocols. In one implementation, the master gaming controller **1012** communicates with other devices using a serial communication protocol. A few non-limiting examples of serial communication protocols that other devices, such as peripherals (e.g., a bill validator or a ticket printer), may use to communicate with the master game controller **1012** include USB, RS-232, and Netplex (a proprietary protocol developed by IGT).

[0086] As will be appreciated by one skilled in the art, aspects of the present disclosure may be illustrated and described herein in any of a number of patentable classes or context including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, aspects of the present disclosure may be implemented entirely hardware, entirely software (including firmware, resident software, micro-code, etc.) or combining software and hardware implementation that may all generally be referred to herein as a “circuit,” “module,” “component,” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable media having computer readable program code embodied thereon.

[0087] Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C #, VB.NET, Python or the like, conventional procedural programming languages, such as the “C” programming language, Visual Basic, Fortran 2003, Perl, COBOL 2002, PHP, ABAP, dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the user’s computer, partly on the user’s computer, as a stand-alone software package, partly on the user’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user’s computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (SaaS).

[0088] Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other program-

mable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0089] These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0090] In certain embodiments, the at least one memory device **1016** is configured to store program code and instructions executable by the at least one processor of the EGM to control the EGM. The at least one memory device **1016** of the EGM also stores other operating data, such as image data, event data, input data, random number generators (RNGs) or pseudo-RNGs, payable data or information, and/or applicable game rules that relate to the play of one or more games on the EGM. In various embodiments, part or all of the program code and/or the operating data described above is stored in at least one detachable or removable memory device including, but not limited to, a cartridge, a disk, a CD ROM, a DVD, a USB memory device, or any other suitable non-transitory computer readable medium. In certain such embodiments, an operator (such as a gaming establishment operator) and/or a player uses such a removable memory device in an EGM to implement at least part of the present disclosure. In other embodiments, part or all of the program code and/or the operating data is downloaded to the at least one memory device of the EGM through any suitable data network described above (such as an Internet or intranet).

[0091] The at least one memory device **1016** also stores a plurality of device drivers **1042**. Examples of different types of device drivers include device drivers for EGM components and device drivers for the peripheral components **1022**. Typically, the device drivers **1042** utilize various communication protocols that enable communication with a particular physical device. The device driver abstracts the hardware implementation of that device. For example, a device driver may be written for each type of card reader that could potentially be connected to the EGM. Non-limiting examples of communication protocols used to implement the device drivers include Netplex, USB, Serial, Ethernet **175**, Firewire, I/O debouncer, direct memory map, serial, PCI, parallel, RF, Bluetooth™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), etc. In one embodiment, when one type of a particular device is exchanged for another type of the particular device, the at least one processor of the EGM loads the new device driver from the at least one memory device to enable communication with the new device. For instance, one type of card reader in the EGM can be replaced with a second different

type of card reader when device drivers for both card readers are stored in the at least one memory device.

[0092] In certain embodiments, the software units stored in the at least one memory device **1016** can be upgraded as needed. For instance, when the at least one memory device **1016** is a hard drive, new games, new game options, new parameters, new settings for existing parameters, new settings for new parameters, new device drivers, and new communication protocols can be uploaded to the at least one memory device **1016** from the master game controller **1012** or from some other external device. As another example, when the at least one memory device **1016** includes a CD/DVD drive including a CD/DVD configured to store game options, parameters, and settings, the software stored in the at least one memory device **1016** can be upgraded by replacing a first CD/DVD with a second CD/DVD. In yet another example, when the at least one memory device **1016** uses flash memory **1019** or EPROM **1008** units configured to store games, game options, parameters, and settings, the software stored in the flash and/or EPROM memory units can be upgraded by replacing one or more memory units with new memory units that include the upgraded software. In another embodiment, one or more of the memory devices, such as the hard drive, may be employed in a game software download process from a remote software server.

[0093] In some embodiments, the at least one memory device **1016** also stores authentication and/or validation components **1044** configured to authenticate/validate specified EGM components and/or information, such as hardware components, software components, firmware components, peripheral device components, user input device components, information received from one or more user input devices, information stored in the at least one memory device **1016**, etc.

[0094] In certain embodiments, the peripheral devices **1022** include several device interfaces, such as: (1) at least one output device **1020** including at least one display device **1035**; (2) at least one input device **1030** (which may include contact and/or non-contact interfaces); (3) at least one transponder **1054**; (4) at least one wireless communication component **1056**; (5) at least one wired/wireless power distribution component **1058**; (6) at least one sensor **1060**; (7) at least one data preservation component **1062**; (8) at least one motion/gesture analysis and interpretation component **1064**; (9) at least one motion detection component **1066**; (10) at least one portable power source **1068**; (11) at least one geolocation module **1076**; (12) at least one user identification module **1077**; (13) at least one player/device tracking module **1078**; and (14) at least one information filtering module **1079**.

[0095] The at least one output device **1020** includes at least one display device **1035** configured to display any game(s) displayed by the EGM and any suitable information associated with such game(s). In certain embodiments, the display devices are connected to or mounted on a housing of the EGM (described below). In various embodiments, the display devices serve as digital glass configured to advertise certain games or other aspects of the gaming establishment in which the EGM is located. In various embodiments, the EGM includes one or more of the following display devices: (a) a central display device; (b) a player tracking display configured to display various information regarding a player's player tracking status (as described below); (c) a secondary or upper display device in addition to the central

display device and the player tracking display; (d) a credit display configured to display a current quantity of credits, amount of cash, account balance, or the equivalent; and (e) a bet display configured to display an amount wagered for one or more plays of one or more games. The example EGM **2000a** illustrated in FIG. 4A includes a central display device **2116**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**. The example EGM **2000b** illustrated in FIG. 4B includes a central display device **2116**, an upper display device **2118**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**.

[0096] In various embodiments, the display devices include, without limitation: a monitor, a television display, a plasma display, a liquid crystal display (LCD), a display based on light emitting diodes (LEDs), a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display device includes a touch-screen with an associated touch-screen controller. The display devices may be of any suitable sizes, shapes, and configurations.

[0097] The display devices of the EGM are configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices of the EGM are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices of the EGM are configured to display one or more video reels, one or more video wheels, and/or one or more video dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

[0098] In various embodiments, the at least one output device **1020** includes a payout device. In these embodiments, after the EGM receives an actuation of a cashout device (described below), the EGM causes the payout device to provide a payment to the player. In one embodiment, the payout device is one or more of: (a) a ticket printer and dispenser configured to print and dispense a ticket or credit slip associated with a monetary value, wherein the ticket or credit slip may be redeemed for its monetary value via a cashier, a kiosk, or other suitable redemption system; (b) a bill dispenser configured to dispense paper currency; (c) a coin dispenser configured to dispense coins or tokens (such as into a coin payout tray); and (d) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a ticket printer and dispenser **2136**.

[0099] In certain embodiments, rather than dispensing bills, coins, or a physical ticket having a monetary value to the player following receipt of an actuation of the cashout device, the payout device is configured to cause a payment to be provided to the player in the form of an electronic funds transfer, such as via a direct deposit into a bank account, a casino account, or a prepaid account of the player;

via a transfer of funds onto an electronically recordable identification card or smart card of the player; or via sending a virtual ticket having a monetary value to an electronic device of the player.

[0100] While any credit balances, any wagers, any values, and any awards are described herein as amounts of monetary credits or currency, one or more of such credit balances, such wagers, such values, and such awards may be for non-monetary credits, promotional credits, of player tracking points or credits.

[0101] In certain embodiments, the at least one output device **1020** is a sound generating device controlled by one or more sound cards. In one such embodiment, the sound generating device includes one or more speakers or other sound generating hardware and/or software configured to generate sounds, such as by playing music for any games or by playing music for other modes of the EGM, such as an attract mode. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a plurality of speakers **2150**. In another such embodiment, the EGM provides dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the EGM. In certain embodiments, the EGM displays a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the EGM. The videos may be customized to provide any appropriate information.

[0102] The at least one input device **1030** may include any suitable device that enables an input signal to be produced and received by the at least one processor **1010** of the EGM.

[0103] In one embodiment, the at least one input device **1030** includes a payment device configured to communicate with the at least one processor of the EGM to fund the EGM. In certain embodiments, the payment device includes one or more of: (a) a bill acceptor into which paper money is inserted to fund the EGM; (b) a ticket acceptor into which a ticket or a voucher is inserted to fund the EGM; (c) a coin slot into which coins or tokens are inserted to fund the EGM; (d) a reader or a validator for credit cards, debit cards, or credit slips into which a credit card, debit card, or credit slip is inserted to fund the EGM; (e) a player identification card reader into which a player identification card is inserted to fund the EGM; or (f) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a combined bill and ticket acceptor **2128** and a coin slot **2126**.

[0104] In one embodiment, the at least one input device **1030** includes a payment device configured to enable the EGM to be funded via an electronic funds transfer, such as a transfer of funds from a bank account. In another embodiment, the EGM includes a payment device configured to communicate with a mobile device of a player, such as a mobile phone, a radio frequency identification tag, or any other suitable wired or wireless device, to retrieve relevant information associated with that player to fund the EGM. When the EGM is funded, the at least one processor determines the amount of funds entered and displays the corresponding amount on a credit display or any other suitable display as described below.

[0105] In certain embodiments, the at least one input device **1030** includes at least one wagering or betting device. In various embodiments, the one or more wagering or

betting devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). One such wagering or betting device is as a maximum wager or bet device that, when actuated, causes the EGM to place a maximum wager on a play of a game. Another such wagering or betting device is a repeat bet device that, when actuated, causes the EGM to place a wager that is equal to the previously-placed wager on a play of a game. A further such wagering or betting device is a bet one device that, when actuated, causes the EGM to increase the wager by one credit. Generally, upon actuation of one of the wagering or betting devices, the quantity of credits displayed in a credit meter (described below) decreases by the amount of credits wagered, while the quantity of credits displayed in a bet display (described below) increases by the amount of credits wagered.

[0106] In various embodiments, the at least one input device **1030** includes at least one game play activation device. In various embodiments, the one or more game play initiation devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). After a player appropriately funds the EGM and places a wager, the EGM activates the game play activation device to enable the player to actuate the game play activation device to initiate a play of a game on the EGM (or another suitable sequence of events associated with the EGM). After the EGM receives an actuation of the game play activation device, the EGM initiates the play of the game. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a game play activation device in the form of a game play initiation button **2132**. In other embodiments, the EGM begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

[0107] In other embodiments, the at least one input device **1030** includes a cashout device. In various embodiments, the cashout device is: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). When the EGM receives an actuation of the cashout device from a player and the player has a positive (i.e., greater-than-zero) credit balance, the EGM initiates a payout associated with the player's credit balance. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a cashout device in the form of a cashout button **2134**.

[0108] In various embodiments, the at least one input device **1030** includes a plurality of buttons that are programmable by the EGM operator to, when actuated, cause the EGM to perform particular functions. For instance, such buttons may be hard keys, programmable soft keys, or icons icon displayed on a display device of the EGM (described below) that are actuable via a touch screen of the EGM (described below) or via use of a suitable input device of the

EGM (such as a mouse or a joystick). The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a plurality of such buttons **2130**.

[0109] In certain embodiments, the at least one input device **1030** includes a touch-screen coupled to a touch-screen controller or other touch-sensitive display overlay to enable interaction with any images displayed on a display device (as described below). One such input device is a conventional touch-screen button panel. The touch-screen and the touch-screen controller are connected to a video controller. In these embodiments, signals are input to the EGM by touching the touch screen at the appropriate locations.

[0110] In embodiments including a player tracking system, as further described below, the at least one input device **1030** includes a card reader in communication with the at least one processor of the EGM. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a card reader **2138**. The card reader is configured to read a player identification card inserted into the card reader.

[0111] The at least one wireless communication component **1056** includes one or more communication interfaces having different architectures and utilizing a variety of protocols, such as (but not limited to) 802.11 (WiFi); 802.15 (including Bluetooth™); 802.16 (WiMax); 802.22; cellular standards such as CDMA, CDMA2000, and WCDMA; Radio Frequency (e.g., RFID); infrared; and Near Field Magnetic communication protocols. The at least one wireless communication component **1056** transmits electrical, electromagnetic, or optical signals that carry digital data streams or analog signals representing various types of information.

[0112] The at least one wired/wireless power distribution component **1058** includes components or devices that are configured to provide power to other devices. For example, in one embodiment, the at least one power distribution component **1058** includes a magnetic induction system that is configured to provide wireless power to one or more user input devices near the EGM. In one embodiment, a user input device docking region is provided, and includes a power distribution component that is configured to recharge a user input device without requiring metal-to-metal contact. In one embodiment, the at least one power distribution component **1058** is configured to distribute power to one or more internal components of the EGM, such as one or more rechargeable power sources (e.g., rechargeable batteries) located at the EGM.

[0113] In certain embodiments, the at least one sensor **1060** includes at least one of: optical sensors, pressure sensors, RF sensors, infrared sensors, image sensors, thermal sensors, and biometric sensors. The at least one sensor **1060** may be used for a variety of functions, such as: detecting movements and/or gestures of various objects within a predetermined proximity to the EGM; detecting the presence and/or identity of various persons (e.g., players, casino employees, etc.), devices (e.g., user input devices), and/or systems within a predetermined proximity to the EGM.

[0114] The at least one data preservation component **1062** is configured to detect or sense one or more events and/or conditions that, for example, may result in damage to the EGM and/or that may result in loss of information associated with the EGM. Additionally, the data preservation

system **1062** may be operable to initiate one or more appropriate action(s) in response to the detection of such events/conditions.

[0115] The at least one motion/gesture analysis and interpretation component **1064** is configured to analyze and/or interpret information relating to detected player movements and/or gestures to determine appropriate player input information relating to the detected player movements and/or gestures. For example, in one embodiment, the at least one motion/gesture analysis and interpretation component **1064** is configured to perform one or more of the following functions: analyze the detected gross motion or gestures of a player; interpret the player's motion or gestures (e.g., in the context of a casino game being played) to identify instructions or input from the player; utilize the interpreted instructions/input to advance the game state; etc. In other embodiments, at least a portion of these additional functions may be implemented at a remote system or device.

[0116] The at least one portable power source **1068** enables the EGM to operate in a mobile environment. For example, in one embodiment, the EGM **300** includes one or more rechargeable batteries.

[0117] The at least one geolocation module **1076** is configured to acquire geolocation information from one or more remote sources and use the acquired geolocation information to determine information relating to a relative and/or absolute position of the EGM. For example, in one implementation, the at least one geolocation module **1076** is configured to receive GPS signal information for use in determining the position or location of the EGM. In another implementation, the at least one geolocation module **1076** is configured to receive multiple wireless signals from multiple remote devices (e.g., EGMs, servers, wireless access points, etc.) and use the signal information to compute position/location information relating to the position or location of the EGM.

[0118] The at least one user identification module **1077** is configured to determine the identity of the current user or current owner of the EGM. For example, in one embodiment, the current user is required to perform a login process at the EGM in order to access one or more features. Alternatively, the EGM is configured to automatically determine the identity of the current user based on one or more external signals, such as an RFID tag or badge worn by the current user and that provides a wireless signal to the EGM that is used to determine the identity of the current user. In at least one embodiment, various security features are incorporated into the EGM to prevent unauthorized users from accessing confidential or sensitive information.

[0119] The at least one information filtering module **1079** is configured to perform filtering (e.g., based on specified criteria) of selected information to be displayed at one or more displays **1035** of the EGM.

[0120] In various embodiments, the EGM includes a plurality of communication ports configured to enable the at least one processor of the EGM to communicate with and to operate with external peripherals, such as: accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, SCSI ports, sole-

noids, speakers, thumbsticks, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices.

[0121] As generally described above, in certain embodiments, such as the example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B**, the EGM has a support structure, housing, or cabinet that provides support for a plurality of the input devices and the output devices of the EGM. Further, the EGM is configured such that a player may operate it while standing or sitting. In various embodiments, the EGM is positioned on a base or stand, or is configured as a pub-style tabletop game (not shown) that a player may operate typically while sitting. As illustrated by the different example EGMs **2000a** and **2000b** shown in FIGS. **4A** and **4B**, EGMs may have varying housing and display configurations.

[0122] In certain embodiments, the EGM is a device that has obtained approval from a regulatory gaming commission, and in other embodiments, the EGM is a device that has not obtained approval from a regulatory gaming commission.

[0123] The EGMs described above are merely three examples of different types of EGMs. Certain of these example EGMs may include one or more elements that may not be included in all gaming systems, and these example EGMs may not include one or more elements that are included in other gaming systems. For example, certain EGMs include a coin acceptor while others do not.

[0124] In various embodiments, an EGM may be implemented in one of a variety of different configurations. In various embodiments, the EGM may be implemented as one of: (a) a dedicated EGM in which computerized game programs executable by the EGM for controlling any primary or base games (referred to herein as “primary games”) and/or any secondary or bonus games or other functions (referred to herein as “secondary games”) displayed by the EGM are provided with the EGM before delivery to a gaming establishment or before being provided to a player; and (b) a changeable EGM in which computerized game programs executable by the EGM for controlling any primary games and/or secondary games displayed by the EGM are downloadable or otherwise transferred to the EGM through a data network or remote communication link; from a USB drive, flash memory card, or other suitable memory device; or in any other suitable manner after the EGM is physically located in a gaming establishment or after the EGM is provided to a player.

[0125] As generally explained above, in various embodiments in which the gaming system includes a server and a changeable EGM, the at least one memory device of the server stores different game programs and instructions executable by the at least one processor of the changeable EGM to control one or more primary games and/or secondary games displayed by the changeable EGM. More specifically, each such executable game program represents a different game or a different type of game that the at least one changeable EGM is configured to operate. In one example, certain of the game programs are executable by the changeable EGM to operate games having the same or substantially the same game play but different paytables. In different embodiments, each executable game program is associated with a primary game, a secondary game, or both. In certain embodiments, an executable game program is executable by the at least one processor of the at least one

changeable EGM as a secondary game to be played simultaneously with a play of a primary game (which may be downloaded to or otherwise stored on the at least one changeable EGM), or vice versa.

[0126] In operation of such embodiments, the server is configured to communicate one or more of the stored executable game programs to the at least one processor of the changeable EGM. In different embodiments, a stored executable game program is communicated or delivered to the at least one processor of the changeable EGM by: (a) embedding the executable game program in a device or a component (such as a microchip to be inserted into the changeable EGM); (b) writing the executable game program onto a disc or other media; or (c) uploading or streaming the executable game program over a data network (such as a dedicated data network). After the executable game program is communicated from the server to the changeable EGM, the at least one processor of the changeable EGM executes the executable game program to enable the primary game and/or the secondary game associated with that executable game program to be played using the display device(s) and/or the input device(s) of the changeable EGM. That is, when an executable game program is communicated to the at least one processor of the changeable EGM, the at least one processor of the changeable EGM changes the game or the type of game that may be played using the changeable EGM.

[0127] In certain embodiments, the gaming system randomly determines any game outcome(s) (such as a win outcome) and/or award(s) (such as a quantity of credits to award for the win outcome) for a play of a primary game and/or a play of a secondary game based on probability data. In certain such embodiments, this random determination is provided through utilization of an RNG, such as a true RNG or a pseudo RNG, or any other suitable randomization process. In one such embodiment, each game outcome or award is associated with a probability, and the gaming system generates the game outcome(s) and/or the award(s) to be provided based on the associated probabilities. In these embodiments, since the gaming system generates game outcomes and/or awards randomly or based on one or more probability calculations, there is no certainty that the gaming system will ever provide any specific game outcome and/or award.

[0128] In certain embodiments, the gaming system maintains one or more predetermined pools or sets of predetermined game outcomes and/or awards. In certain such embodiments, upon generation or receipt of a game outcome and/or award request, the gaming system independently selects one of the predetermined game outcomes and/or awards from the one or more pools or sets. The gaming system flags or marks the selected game outcome and/or award as used. Once a game outcome or an award is flagged as used, it is prevented from further selection from its respective pool or set; that is, the gaming system does not select that game outcome or award upon another game outcome and/or award request. The gaming system provides the selected game outcome and/or award.

[0129] In certain embodiments, the gaming system determines a predetermined game outcome and/or award based on the results of a bingo, keno, or lottery game. In certain such embodiments, the gaming system utilizes one or more bingo, keno, or lottery games to determine the predetermined game outcome and/or award provided for a primary

game and/or a secondary game. The gaming system is provided or associated with a bingo card. Each bingo card consists of a matrix or array of elements, wherein each element is designated with separate indicia. After a bingo card is provided, the gaming system randomly selects or draws a plurality of the elements. As each element is selected, a determination is made as to whether the selected element is present on the bingo card. If the selected element is present on the bingo card, that selected element on the provided bingo card is marked or flagged. This process of selecting elements and marking any selected elements on the provided bingo cards continues until one or more predetermined patterns are marked on one or more of the provided bingo cards. After one or more predetermined patterns are marked on one or more of the provided bingo cards, game outcome and/or award is determined based, at least in part, on the selected elements on the provided bingo cards.

[0130] In certain embodiments in which the gaming system includes a server and an EGM, the EGM is configured to communicate with the server for monitoring purposes only. In such embodiments, the EGM determines the game outcome(s) and/or award(s) to be provided in any of the manners described above, and the server monitors the activities and events occurring on the EGM. In one such embodiment, the gaming system includes a real-time or online accounting and gaming information system configured to communicate with the server. In this embodiment, the accounting and gaming information system includes: (a) a player database configured to store player profiles, (b) a player tracking module configured to track players (as described below), and (c) a credit system configured to provide automated transactions.

[0131] As noted above, in various embodiments, the gaming system includes one or more executable game programs executable by at least one processor of the gaming system to provide one or more primary games and one or more secondary games. The primary game(s) and the secondary game(s) may comprise any suitable games and/or wagering games, such as, but not limited to: electro-mechanical or video slot or spinning reel type games; video card games such as video draw poker, multi-hand video draw poker, other video poker games, video blackjack games, and video baccarat games; video keno games; video bingo games; and video selection games.

[0132] In certain embodiments in which the primary game is a slot or spinning reel type game, the gaming system includes one or more reels in either an electromechanical form with mechanical rotating reels or in a video form with simulated reels and movement thereof. Each reel displays a plurality of indicia or symbols, such as bells, hearts, fruits, numbers, letters, bars, or other images that typically correspond to a theme associated with the gaming system. In certain such embodiments, the gaming system includes one or more paylines associated with the reels. The example EGM 2000b shown in FIG. 4B includes a payline 2152 and a plurality of reels 2154. In certain embodiments, one or more of the reels are independent reels or unisymbol reels. In such embodiments, each independent reel generates and displays one symbol.

[0133] In various embodiments, one or more of the paylines is horizontal, vertical, circular, diagonal, angled, or any suitable combination thereof. In other embodiments, each of one or more of the paylines is associated with a plurality of adjacent symbol display areas on a requisite number of

adjacent reels. In one such embodiment, one or more paylines are formed between at least two symbol display areas that are adjacent to each other by either sharing a common side or sharing a common corner (i.e., such paylines are connected paylines). The gaming system enables a wager to be placed on one or more of such paylines to activate such paylines. In other embodiments in which one or more paylines are formed between at least two adjacent symbol display areas, the gaming system enables a wager to be placed on a plurality of symbol display areas, which activates those symbol display areas.

[0134] In various embodiments, the gaming system provides one or more awards after a spin of the reels when specified types and/or configurations of the indicia or symbols on the reels occur on an active payline or otherwise occur in a winning pattern, occur on the requisite number of adjacent reels, and/or occur in a scatter pay arrangement.

[0135] In certain embodiments, the gaming system employs a way to win award determination. In these embodiments, any outcome to be provided is determined based on a number of associated symbols that are generated in active symbol display areas on the requisite number of adjacent reels (i.e., not on paylines passing through any displayed winning symbol combinations). If a winning symbol combination is generated on the reels, one award for that occurrence of the generated winning symbol combination is provided.

[0136] In various embodiments, the gaming system includes a progressive award. Typically, a progressive award includes an initial amount and an additional amount funded through a portion of each wager placed to initiate a play of a primary game. When one or more triggering events occurs, the gaming system provides at least a portion of the progressive award. After the gaming system provides the progressive award, an amount of the progressive award is reset to the initial amount and a portion of each subsequent wager is allocated to the next progressive award.

[0137] As generally noted above, in addition to providing winning credits or other awards for one or more plays of the primary game(s), in various embodiments the gaming system provides credits or other awards for one or more plays of one or more secondary games. The secondary game typically enables an award to be obtained in addition to any award obtained through play of the primary game(s). The secondary game(s) typically produces a higher level of player excitement than the primary game(s) because the secondary game(s) provides a greater expectation of winning than the primary game(s) and is accompanied with more attractive or unusual features than the primary game(s). The secondary game(s) may be any type of suitable game, either similar to or completely different from the primary game.

[0138] In various embodiments, the gaming system automatically provides or initiates the secondary game upon the occurrence of a triggering event or the satisfaction of a qualifying condition. In other embodiments, the gaming system initiates the secondary game upon the occurrence of the triggering event or the satisfaction of the qualifying condition and upon receipt of an initiation input. In certain embodiments, the triggering event or qualifying condition is a selected outcome in the primary game(s) or a particular arrangement of one or more indicia on a display device for a play of the primary game(s), such as a “BONUS” symbol appearing on three adjacent reels along a payline following

a spin of the reels for a play of the primary game. In other embodiments, the triggering event or qualifying condition occurs based on a certain amount of game play (such as number of games, number of credits, amount of time) being exceeded, or based on a specified number of points being earned during game play. Any suitable triggering event or qualifying condition or any suitable combination of a plurality of different triggering events or qualifying conditions may be employed.

[0139] In other embodiments, at least one processor of the gaming system randomly determines when to provide one or more plays of one or more secondary games. In one such embodiment, no apparent reason is provided for providing the secondary game. In this embodiment, qualifying for a secondary game is not triggered by the occurrence of an event in any primary game or based specifically on any of the plays of any primary game. That is, qualification is provided without any explanation or, alternatively, with a simple explanation. In another such embodiment, the gaming system determines qualification for a secondary game at least partially based on a game triggered or symbol triggered event, such as at least partially based on play of a primary game.

[0140] In various embodiments, after qualification for a secondary game has been determined, the secondary game participation may be enhanced through continued play on the primary game. Thus, in certain embodiments, for each secondary game qualifying event, such as a secondary game symbol, that is obtained, a given number of secondary game wagering points or credits is accumulated in a “secondary game meter” configured to accrue the secondary game wagering credits or entries toward eventual participation in the secondary game. In one such embodiment, the occurrence of multiple such secondary game qualifying events in the primary game results in an arithmetic or exponential increase in the number of secondary game wagering credits awarded. In another such embodiment, any extra secondary game wagering credits may be redeemed during the secondary game to extend play of the secondary game.

[0141] In certain embodiments, no separate entry fee or buy-in for the secondary game is required. That is, entry into the secondary game cannot be purchased; rather, in these embodiments entry must be won or earned through play of the primary game, thereby encouraging play of the primary game. In other embodiments, qualification for the secondary game is accomplished through a simple “buy-in.” For example, qualification through other specified activities is unsuccessful, payment of a fee or placement of an additional wager “buys-in” to the secondary game. In certain embodiments, a separate side wager must be placed on the secondary game or a wager of a designated amount must be placed on the primary game to enable qualification for the secondary game. In these embodiments, the secondary game triggering event must occur and the side wager (or designated primary game wager amount) must have been placed for the secondary game to trigger.

[0142] In various embodiments in which the gaming system includes a plurality of EGMs, the EGMs are configured to communicate with one another to provide a group gaming environment. In certain such embodiments, the EGMs enable players of those EGMs to work in conjunction with one another, such as by enabling the players to play together as a team or group, to win one or more awards. In other such embodiments, the EGMs enable players of those EGMs to

compete against one another for one or more awards. In one such embodiment, the EGMs enable the players of those EGMs to participate in one or more gaming tournaments for one or more awards.

[0143] In various embodiments, the gaming system includes one or more player tracking systems. Such player tracking systems enable operators of the gaming system (such as casinos or other gaming establishments) to recognize the value of customer loyalty by identifying frequent customers and rewarding them for their patronage. Such a player tracking system is configured to track a player’s gaming activity. In one such embodiment, the player tracking system does so through the use of player tracking cards. In this embodiment, a player is issued a player identification card that has an encoded player identification number that uniquely identifies the player. When the player’s playing tracking card is inserted into a card reader of the gaming system to begin a gaming session, the card reader reads the player identification number off the player tracking card to identify the player. The gaming system timely tracks any suitable information or data relating to the identified player’s gaming session. The gaming system also timely tracks when the player tracking card is removed to conclude play for that gaming session. In another embodiment, rather than requiring insertion of a player tracking card into the card reader, the gaming system utilizes one or more portable devices, such as a mobile phone, a radio frequency identification tag, or any other suitable wireless device, to track when a gaming session begins and ends. In another embodiment, the gaming system utilizes any suitable biometric technology or ticket technology to track when a gaming session begins and ends.

[0144] In such embodiments, during one or more gaming sessions, the gaming system tracks any suitable information or data, such as any amounts wagered, average wager amounts, and/or the time at which these wagers are placed. In different embodiments, for one or more players, the player tracking system includes the player’s account number, the player’s card number, the player’s first name, the player’s surname, the player’s preferred name, the player’s player tracking ranking, any promotion status associated with the player’s player tracking card, the player’s address, the player’s birthday, the player’s anniversary, the player’s recent gaming sessions, or any other suitable data. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed on a player tracking display. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed via one or more service windows that are displayed on the central display device and/or the upper display device.

[0145] In various embodiments, the gaming system includes one or more servers configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable web-based game play using the personal gaming device. In various embodiments, the player must first access a gaming website via an Internet browser of the personal gaming device or execute an application (commonly called an “app”) installed on the personal gaming device before the player can use the personal gaming device to participate in web-based game play. In certain embodiments, the one or more servers and the personal gaming device operate in a thin-client environment. In these embodiments, the personal gaming device receives inputs via one or more input devices

(such as a touch screen and/or physical buttons), the personal gaming device sends the received inputs to the one or more servers, the one or more servers make various determinations based on the inputs and determine content to be displayed (such as a randomly determined game outcome and corresponding award), the one or more servers send the content to the personal gaming device, and the personal gaming device displays the content.

[0146] In certain such embodiments, the one or more servers must identify the player before enabling game play on the personal gaming device (or, in some embodiments, before enabling monetary wager-based game play on the personal gaming device). In these embodiments, the player must identify herself to the one or more servers, such as by inputting the player's unique username and password combination, providing an input to a biometric sensor (e.g., a fingerprint sensor, a retinal sensor, a voice sensor, or a facial-recognition sensor), or providing any other suitable information.

[0147] Once identified, the one or more servers enable the player to establish an account balance from which the player can draw credits usable to wager on plays of a game. In certain embodiments, the one or more servers enable the player to initiate an electronic funds transfer to transfer funds from a bank account to the player's account balance. In other embodiments, the one or more servers enable the player to make a payment using the player's credit card, debit card, or other suitable device to add money to the player's account balance. In other embodiments, the one or more servers enable the player to add money to the player's account balance via a peer-to-peer type application, such as PayPal or Venmo. The one or more servers also enable the player to cash out the player's account balance (or part of it) in any suitable manner, such as via an electronic funds transfer, by initiating creation of a paper check that is mailed to the player, or by initiating printing of a voucher at a kiosk in a gaming establishment.

[0148] In certain embodiments, the one or more servers include a payment server that handles establishing and cashing out players' account balances and a separate game server configured to determine the outcome and any associated award for a play of a game. In these embodiments, the game server is configured to communicate with the personal gaming device and the payment device, and the personal gaming device and the payment device are not configured to directly communicate with one another. In these embodiments, when the game server receives data representing a request to start a play of a game at a desired wager, the game server sends data representing the desired wager to the payment server. The payment server determines whether the player's account balance can cover the desired wager (i.e., includes a monetary balance at least equal to the desired wager).

[0149] If the payment server determines that the player's account balance cannot cover the desired wager, the payment server notifies the game server, which then instructs the personal gaming device to display a suitable notification to the player that the player's account balance is too low to place the desired wager. If the payment server determines that the player's account balance can cover the desired wager, the payment server deducts the desired wager from the account balance and notifies the game server. The game server then determines an outcome and any associated award for the play of the game. The game server notifies the

payment server of any nonzero award, and the payment server increases the player's account balance by the nonzero award. The game server sends data representing the outcome and any award to the personal gaming device, which displays the outcome and any award.

[0150] In certain embodiments, the one or more servers enable web-based game play using a personal gaming device only if the personal gaming device satisfies one or more jurisdictional requirements. In one embodiment, the one or more servers enable web-based game play using the personal gaming device only if the personal gaming device is located within a designated geographic area (such as within certain state or county lines or within the boundaries of a gaming establishment). In this embodiment, the geolocation module of the personal gaming device determines the location of the personal gaming device and sends the location to the one or more servers, which determine whether the personal gaming device is located within the designated geographic area. In various embodiments, the one or more servers enable non-monetary wager-based game play if the personal gaming device is located outside of the designated geographic area.

[0151] In various embodiments, the gaming system includes an EGM configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable tethered mobile game play using the personal gaming device. Generally, in these embodiments, the EGM establishes communication with the personal gaming device and enables the player to play games on the EGM remotely via the personal gaming device. In certain embodiments, the gaming system includes a geo-fence system that enables tethered game play within a particular geographic area but not outside of that geographic area.

[0152] In certain embodiments, the gaming system is configured to communicate with a social network server that hosts or partially hosts a social networking website via a data network (such as the Internet) to integrate a player's gaming experience with the player's social networking account. This enables the gaming system to send certain information to the social network server that the social network server can use to create content (such as text, an image, and/or a video) and post it to the player's wall, newsfeed, or similar area of the social networking website accessible by the player's connections (and in certain cases the public) such that the player's connections can view that information. This also enables the gaming system to receive certain information from the social network server, such as the player's likes or dislikes or the player's list of connections. In certain embodiments, the gaming system enables the player to link the player's player account to the player's social networking account(s). This enables the gaming system to, once it identifies the player and initiates a gaming session (such as via the player logging in to a website (or an application) on the player's personal gaming device or via the player inserting the player's player tracking card into an EGM), link that gaming session to the player's social networking account(s). In other embodiments, the gaming system enables the player to link the player's social networking account(s) to individual gaming sessions when desired by providing the required login information.

[0153] For instance, in one embodiment, if a player wins a particular award (e.g., a progressive award or a jackpot award) or an award that exceeds a certain threshold (e.g., an

award exceeding \$1,000), the gaming system sends information about the award to the social network server to enable the server to create associated content (such as a screenshot of the outcome and associated award) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to play). In another embodiment, if a player joins a multiplayer game and there is another seat available, the gaming system sends that information to the social network server to enable the server to create associated content (such as text indicating a vacancy for that particular game) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to fill the vacancy). In another embodiment, if the player consents, the gaming system sends advertisement information or offer information to the social network server to enable the social network server to create associated content (such as text or an image reflecting an advertisement and/or an offer) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see. In another embodiment, the gaming system enables the player to recommend a game to the player's connections by posting a recommendation to the player's wall (or other suitable area) of the social networking website.

[0154] Certain of the gaming systems described herein, such as EGMs located in a casino or another gaming establishment, include certain components and/or are configured to operate in certain manners that differentiate these systems from general purpose computing devices, i.e., certain personal gaming devices such as desktop computers and laptop computers.

[0155] For instance, EGMs are highly regulated to ensure fairness and, in many cases, EGMs are configured to award monetary awards up to multiple millions of dollars. To satisfy security and regulatory requirements in a gaming environment, hardware and/or software architectures are implemented in EGMs that differ significantly from those of general purpose computing devices. For purposes of illustration, a description of EGMs relative to general purpose computing devices and some examples of these additional (or different) hardware and/or software architectures found in EGMs are described below.

[0156] At first glance, one might think that adapting general purpose computing device technologies to the gaming industry and EGMs would be a simple proposition because both general purpose computing devices and EGMs employ processors that control a variety of devices. However, due to at least: (1) the regulatory requirements placed on EGMs, (2) the harsh environment in which EGMs operate, (3) security requirements, and (4) fault tolerance requirements, adapting general purpose computing device technologies to EGMs can be quite difficult. Further, techniques and methods for solving a problem in the general purpose computing device industry, such as device compatibility and connectivity issues, might not be adequate in the gaming industry. For instance, a fault or a weakness tolerated in a general purpose computing device, such as security holes in software or frequent crashes, is not tolerated in an EGM because in an EGM these faults can lead to a direct loss of funds from the EGM, such as stolen cash or loss of revenue when the EGM is not operating properly or when the random outcome determination is manipulated.

[0157] Certain differences between general purpose computing devices and EGMs are described below. A first difference between EGMs and general purpose computing devices is that EGMs are state-based systems. A state-based system stores and maintains its current state in a non-volatile memory such that, in the event of a power failure or other malfunction, the state-based system can return to that state when the power is restored or the malfunction is remedied. For instance, for a state-based EGM, if the EGM displays an award for a game of chance but the power to the EGM fails before the EGM provides the award to the player, the EGM stores the pre-power failure state in a non-volatile memory, returns to that state upon restoration of power, and provides the award to the player. This requirement affects the software and hardware design on EGMs. General purpose computing devices are not state-based machines, and a majority of data is usually lost when a malfunction occurs on a general purpose computing device.

[0158] A second difference between EGMs and general purpose computing devices is that, for regulatory purposes, the software on the EGM utilized to operate the EGM has been designed to be static and monolithic to prevent cheating by the operator of the EGM. For instance, one solution that has been employed in the gaming industry to prevent cheating and to satisfy regulatory requirements has been to manufacture an EGM that can use a proprietary processor running instructions to provide the game of chance from an EPROM or other form of non-volatile memory. The coding instructions on the EPROM are static (non-changeable) and must be approved by a gaming regulators in a particular jurisdiction and installed in the presence of a person representing the gaming jurisdiction. Any changes to any part of the software required to generate the game of chance, such as adding a new device driver used to operate a device during generation of the game of chance, can require burning a new EPROM approved by the gaming jurisdiction and reinstalling the new EPROM on the EGM in the presence of a gaming regulator. Regardless of whether the EPROM solution is used, to gain approval in most gaming jurisdictions, an EGM must demonstrate sufficient safeguards that prevent an operator or a player of an EGM from manipulating the EGM's hardware and software in a manner that gives him an unfair, and in some cases illegal, advantage.

[0159] A third difference between EGMs and general purpose computing devices is authentication—EGMs storing code are configured to authenticate the code to determine if the code is unaltered before executing the code. If the code has been altered, the EGM prevents the code from being executed. The code authentication requirements in the gaming industry affect both hardware and software designs on EGMs. Certain EGMs use hash functions to authenticate code. For instance, one EGM stores game program code, a hash function, and an authentication hash (which may be encrypted). Before executing the game program code, the EGM hashes the game program code using the hash function to obtain a result hash and compares the result hash to the authentication hash. If the result hash matches the authentication hash, the EGM determines that the game program code is valid and executes the game program code. If the result hash does not match the authentication hash, the EGM determines that the game program code has been altered (i.e., may have been tampered with) and prevents execution of the game program code.

[0160] A fourth difference between EGMs and general purpose computing devices is that EGMs have unique peripheral device requirements that differ from those of a general purpose computing device, such as peripheral device security requirements not usually addressed by general purpose computing devices. For instance, monetary devices, such as coin dispensers, bill validators, and ticket printers and computing devices that are used to govern the input and output of cash or other items having monetary value (such as tickets) to and from an EGM have security requirements that are not typically addressed in general purpose computing devices. Therefore, many general purpose computing device techniques and methods developed to facilitate device connectivity and device compatibility do not address the emphasis placed on security in the gaming industry.

[0161] To address some of the issues described above, a number of hardware/software components and architectures are utilized in EGMs that are not typically found in general purpose computing devices. These hardware/software components and architectures, as described below in more detail, include but are not limited to watchdog timers, voltage monitoring systems, state-based software architecture and supporting hardware, specialized communication interfaces, security monitoring, and trusted memory.

[0162] Certain EGMs use a watchdog timer to provide a software failure detection mechanism. In a normally-operating EGM, the operating software periodically accesses control registers in the watchdog timer subsystem to “re-trigger” the watchdog. Should the operating software fail to access the control registers within a preset timeframe, the watchdog timer will timeout and generate a system reset. Typical watchdog timer circuits include a loadable timeout counter register to enable the operating software to set the timeout interval within a certain range of time. A differentiating feature of some circuits is that the operating software cannot completely disable the function of the watchdog timer. In other words, the watchdog timer always functions from the time power is applied to the board.

[0163] Certain EGMs use several power supply voltages to operate portions of the computer circuitry. These can be generated in a central power supply or locally on the computer board. If any of these voltages falls out of the tolerance limits of the circuitry they power, unpredictable operation of the EGM may result. Though most modern general purpose computing devices include voltage monitoring circuitry, these types of circuits only report voltage status to the operating software. Out of tolerance voltages can cause software malfunction, creating a potential uncontrolled condition in the general purpose computing device. Certain EGMs have power supplies with relatively tighter voltage margins than that required by the operating circuitry. In addition, the voltage monitoring circuitry implemented in certain EGMs typically has two thresholds of control. The first threshold generates a software event that can be detected by the operating software and an error condition then generated. This threshold is triggered when a power supply voltage falls out of the tolerance range of the power supply, but is still within the operating range of the circuitry. The second threshold is set when a power supply voltage falls out of the operating tolerance of the circuitry. In this case, the circuitry generates a reset, halting operation of the EGM.

[0164] As described above, certain EGMs are state-based machines. Different functions of the game provided by the

EGM (e.g., bet, play, result, points in the graphical presentation, etc.) may be defined as a state. When the EGM moves a game from one state to another, the EGM stores critical data regarding the game software in a custom non-volatile memory subsystem. This ensures that the player’s wager and credits are preserved and to minimize potential disputes in the event of a malfunction on the EGM. In general, the EGM does not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been stored. This feature enables the EGM to recover operation to the current state of play in the event of a malfunction, loss of power, etc. that occurred just before the malfunction. In at least one embodiment, the EGM is configured to store such critical information using atomic transactions.

[0165] Generally, an atomic operation in computer science refers to a set of operations that can be combined so that they appear to the rest of the system to be a single operation with only two possible outcomes: success or failure. As related to data storage, an atomic transaction may be characterized as series of database operations which either all occur, or all do not occur. A guarantee of atomicity prevents updates to the database occurring only partially, which can result in data corruption.

[0166] To ensure the success of atomic transactions relating to critical information to be stored in the EGM memory before a failure event (e.g., malfunction, loss of power, etc.), memory that includes one or more of the following criteria be used: direct memory access capability; data read/write capability which meets or exceeds minimum read/write access characteristics (such as at least 5.08 Mbytes/sec (Read) and/or at least 38.0 Mbytes/sec (Write)). Memory devices that meet or exceed the above criteria may be referred to as “fault-tolerant” memory devices.

[0167] Typically, battery-backed RAM devices may be configured to function as fault-tolerant devices according to the above criteria, whereas flash RAM and/or disk drive memory are typically not configurable to function as fault-tolerant devices according to the above criteria. Accordingly, battery-backed RAM devices are typically used to preserve EGM critical data, although other types of non-volatile memory devices may be employed. These memory devices are typically not used in typical general purpose computing devices.

[0168] Thus, in at least one embodiment, the EGM is configured to store critical information in fault-tolerant memory (e.g., battery-backed RAM devices) using atomic transactions. Further, in at least one embodiment, the fault-tolerant memory is able to successfully complete all desired atomic transactions (e.g., relating to the storage of EGM critical information) within a time period of 200 milliseconds or less. In at least one embodiment, the time period of 200 milliseconds represents a maximum amount of time for which sufficient power may be available to the various EGM components after a power outage event has occurred at the EGM.

[0169] As described previously, the EGM may not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been atomically stored. After the state of the EGM is restored during the play of a game of chance, game play may resume and the game may be completed in a manner that is no different than if the malfunction had not occurred. Thus, for example, when a malfunction occurs during a game of

chance, the EGM may be restored to a state in the game of chance just before when the malfunction occurred. The restored state may include metering information and graphical information that was displayed on the EGM in the state before the malfunction. For example, when the malfunction occurs during the play of a card game after the cards have been dealt, the EGM may be restored with the cards that were previously displayed as part of the card game. As another example, a bonus game may be triggered during the play of a game of chance in which a player is required to make a number of selections on a video display screen. When a malfunction has occurred after the player has made one or more selections, the EGM may be restored to a state that shows the graphical presentation just before the malfunction including an indication of selections that have already been made by the player. In general, the EGM may be restored to any state in a plurality of states that occur in the game of chance that occurs while the game of chance is played or to states that occur between the play of a game of chance.

[0170] Game history information regarding previous games played such as an amount wagered, the outcome of the game, and the like may also be stored in a non-volatile memory device. The information stored in the non-volatile memory may be detailed enough to reconstruct a portion of the graphical presentation that was previously presented on the EGM and the state of the EGM (e.g., credits) at the time the game of chance was played. The game history information may be utilized in the event of a dispute. For example, a player may decide that in a previous game of chance that they did not receive credit for an award that they believed they won. The game history information may be used to reconstruct the state of the EGM before, during, and/or after the disputed game to demonstrate whether the player was correct or not in the player's assertion.

[0171] Another feature of EGMs is that they often include unique interfaces, including serial interfaces, to connect to specific subsystems internal and external to the EGM. The serial devices may have electrical interface requirements that differ from the "standard" EIA serial interfaces provided by general purpose computing devices. These interfaces may include, for example, Fiber Optic Serial, optically coupled serial interfaces, current loop style serial interfaces, etc. In addition, to conserve serial interfaces internally in the EGM, serial devices may be connected in a shared, daisy-chain fashion in which multiple peripheral devices are connected to a single serial channel.

[0172] The serial interfaces may be used to transmit information using communication protocols that are unique to the gaming industry. For example, IGT's Netplex is a proprietary communication protocol used for serial communication between EGMs. As another example, SAS is a communication protocol used to transmit information, such as metering information, from an EGM to a remote device. Often SAS is used in conjunction with a player tracking system.

[0173] Certain EGMs may alternatively be treated as peripheral devices to a casino communication controller and connected in a shared daisy chain fashion to a single serial interface. In both cases, the peripheral devices are assigned device addresses. If so, the serial controller circuitry must implement a method to generate or detect unique device addresses. General purpose computing device serial ports are not able to do this.

[0174] Security monitoring circuits detect intrusion into an EGM by monitoring security switches attached to access doors in the EGM cabinet. Access violations result in suspension of game play and can trigger additional security operations to preserve the current state of game play. These circuits also function when power is off by use of a battery backup. In power-off operation, these circuits continue to monitor the access doors of the EGM. When power is restored, the EGM can determine whether any security violations occurred while power was off, e.g., via software for reading status registers. This can trigger event log entries and further data authentication operations by the EGM software.

[0175] Trusted memory devices and/or trusted memory sources are included in an EGM to ensure the authenticity of the software that may be stored on less secure memory subsystems, such as mass storage devices. Trusted memory devices and controlling circuitry are typically designed to not enable modification of the code and data stored in the memory device while the memory device is installed in the EGM. The code and data stored in these devices may include authentication algorithms, random number generators, authentication keys, operating system kernels, etc. The purpose of these trusted memory devices is to provide gaming regulatory authorities a root trusted authority within the computing environment of the EGM that can be tracked and verified as original. This may be accomplished via removal of the trusted memory device from the EGM computer and verification of the secure memory device contents is a separate third party verification device. Once the trusted memory device is verified as authentic, and based on the approval of the verification algorithms included in the trusted device, the EGM is enabled to verify the authenticity of additional code and data that may be located in the gaming computer assembly, such as code and data stored on hard disk drives.

[0176] In at least one embodiment, at least a portion of the trusted memory devices/sources may correspond to memory that cannot easily be altered (e.g., "unalterable memory") such as EPROMs, PROMs, Bios, Extended Bios, and/or other memory sources that are able to be configured, verified, and/or authenticated (e.g., for authenticity) in a secure and controlled manner.

[0177] According to one embodiment, when a trusted information source is in communication with a remote device via a network, the remote device may employ a verification scheme to verify the identity of the trusted information source. For example, the trusted information source and the remote device may exchange information using public and private encryption keys to verify each other's identities. In another embodiment, the remote device and the trusted information source may engage in methods using zero knowledge proofs to authenticate each of their respective identities.

[0178] EGMs storing trusted information may utilize apparatuses or methods to detect and prevent tampering. For instance, trusted information stored in a trusted memory device may be encrypted to prevent its misuse. In addition, the trusted memory device may be secured behind a locked door. Further, one or more sensors may be coupled to the memory device to detect tampering with the memory device and provide some record of the tampering. In yet another example, the memory device storing trusted information

might be designed to detect tampering attempts and clear or erase itself when an attempt at tampering has been detected.

[0179] Mass storage devices used in a general purpose computing devices typically enable code and data to be read from and written to the mass storage device. In a gaming environment, modification of the gaming code stored on a mass storage device is strictly controlled and would only be enabled under specific maintenance type events with electronic and physical enablers required. Though this level of security could be provided by software, EGMs that include mass storage devices include hardware level mass storage data protection circuitry that operates at the circuit level to monitor attempts to modify data on the mass storage device and will generate both software and hardware error triggers should a data modification be attempted without the proper electronic and physical enablers being present.

[0180] It should be appreciated that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. For example, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. In another example, the terms “including” and “comprising” and variations thereof, when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. Additionally, a listing of items does not imply that any or all of the items are mutually exclusive nor does a listing of items imply that any or all of the items are collectively exhaustive of anything or in a particular order, unless expressly specified otherwise. Moreover, as used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It should be further appreciated that headings of sections provided in this document and the title are for convenience only, and are not to be taken as limiting the disclosure in any way. Furthermore, unless expressly specified otherwise, devices that are in communication with each other need not be in continuous communication with each other and may communicate directly or indirectly through one or more intermediaries.

[0181] Various changes and modifications to the present embodiments described herein will be apparent to those skilled in the art. For example, a description of an embodiment with several components in communication with each other does not imply that all such components are required, or that each of the disclosed components must communicate with every other component. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present disclosure. As such, these changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended technical scope. It is therefore intended that such changes and modifications be covered by the appended claims.

The invention is claimed as follows:

1. A method of operating a gaming system, the method comprising:

in association with a sequence comprising a plurality of plays of a game:

responsive to the gaming system being associated with a first operational state based on less than a first quantity of points being accumulated, over the plurality of plays of the game, in association with

symbols displayed at a plurality of symbol display positions associated with a first reel set comprising a first plurality of displayed reels, determining, by a processor, a first total award comprising any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the first reel set, wherein the first total award is determined independent of any awards associated with any symbols accumulated at any of a plurality of symbol display positions associated with a second reel set comprising a second plurality of displayed reels,

responsive to the gaming system being associated with a second, different operational state based on at least the first quantity of points being accumulated, over the plurality of plays of the game, in association with symbols displayed at the plurality of symbol display positions associated with the first reel set, determining, by the processor, a second total award comprising any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the first reel set and any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the second reel set, and

causing a display, by a display device, of any determined total award.

2. The method of claim 1, wherein the second total award further comprises an award associated with the second reel set.

3. The method of claim 2, wherein the award associated with the second reel set comprises a progressive award.

4. The method of claim 1, wherein, responsive to at least a first quantity of symbols accumulated, over the plurality of plays of the game, at the plurality of symbol display positions associated with the second reel set, the second total award further comprises a progressive award associated with the second reel set.

5. The method of claim 4, wherein the first quantity of symbols comprises a quantity resulting from a symbol being accumulated, over the plurality of plays of the game, at each of the plurality of symbol display positions associated with the second reel set.

6. The method of claim 1, further comprising, responsive to at least a second quantity of points accumulated, over the plurality of plays of the game, in association with symbols displayed at the plurality of symbol display positions associated with the first reel set and the second reel set, determining, by the processor, a third total award comprising any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the first reel set, any awards associated with any symbols accumulated at any of the plurality of symbol display positions associated with the second reel set, and any awards associated with any symbols accumulated at any of a plurality of symbol display positions associated with a third reel set comprising a third plurality of reels.

7. The method of claim 6, wherein the second quantity of points accumulated, over the plurality of plays of the game, in association with symbols displayed at the plurality of symbol display positions associated with the first reel set and the second reel set is greater than the first quantity of points accumulated, over the plurality of plays of the game, in

association with symbols displayed at the plurality of symbol display positions associated with the first reel set.

8. The method of claim 6, wherein, responsive to at least a first quantity of symbols accumulated, over the plurality of plays of the game, at the plurality of symbol display positions associated with the third reel set, the third total award further comprises a progressive award associated with the third reel set.

9. The method of claim 1, wherein a first symbol display position associated with the first reel set has a first probability of displaying a symbol associated with a first quantity of points and a first symbol display position associated with the second reel set has a second, different probability of displaying a symbol associated with the first quantity of points.

10. A method of operating a gaming system, the method comprising:

in association with a sequence comprising a plurality of plays of a game:

causing a display, by a display device, of a plurality of reel sets, wherein the plurality of reel sets comprises a first reel set in an active state, and a second reel set in an inactive state,

responsive to an occurrence of an accumulation of a first quantity of points associated with the second reel set, modifying, by a processor, an operating state of the gaming system by activating the second reel set from being in the inactive state to being in the active state, and

responsive to an occurrence of a sequence award determination event:

determining, by the processor, a total award, the determination being based on any accumulation of any symbols, over the plurality of plays of the game, at any symbol display positions associated with any of the plurality of reel sets in the active state when the sequence award determination event occurs, and

causing a display, by the display device, of the determined total award.

11. The method of claim 10, wherein, responsive to the second reel set being modified to the active state, the total award further comprises an award associated with the second reel set.

12. The method of claim 11, wherein the award associated with the second reel set comprises a progressive award.

13. The method of claim 10, wherein, responsive to the second reel set being modified to the active state and at least a first quantity of symbols accumulated, over the plurality of plays of the game, at the plurality of symbol display positions associated with the second reel set, the total award further comprises a progressive award associated with the second reel set.

14. The method of claim 10, wherein the modification of the second reel set from being in the inactive state to being in the active state occurs independent of any display of any

points associated with any symbols displayed at any symbol display positions associated with the second reel set in the inactive state.

15. The method of claim 10, wherein the plurality of reel sets comprises a third reel set in the inactive state and further comprising, responsive to an occurrence of an accumulation of a second, different quantity of points associated with the third reel set, modifying, by the processor, the third reel set from being in the inactive state to being in the active state.

16. The gaming system of claim 10, wherein the total award is determined based on any awards associated with any symbols accumulated at any symbol display positions associated with any reel sets in the active state when the sequence award determination event occurs.

17. The method of claim 10, wherein the sequence award determination event comprises a determination that no unplayed games remain associated with the sequence.

18. A method of operating a gaming system, the method comprising:

in association with a sequence comprising a plurality of plays of a game:

causing a display, by a display device, of a plurality of reel sets, wherein:

each reel set comprises a plurality of symbol display positions, and

when the sequence is triggered, each reel set is one of an active reel set and an inactive reel set,

causing a display, by the display device, of an accumulation of symbols, over the plurality of plays of the game, at a quantity of the symbol display positions associated with any of the active reel sets and any of the inactive reel sets,

responsive to an accumulation of a quantity of points associated with any of the inactive reel sets, modifying, by a processor, an operating state of the gaming system by activating that reel set, and

responsive to an occurrence of sequence award determination event:

determining, by the processor, a total award, wherein the total award is based on any symbols accumulated at any of the symbol display positions associated with any of the active reel sets and is independent of any symbols accumulated at any of the symbol display positions associated with any of the inactive reel sets, and

causing a display, by the display device, of the determined total award.

19. The method of claim 18, wherein for each of the active reel sets associated with an occurrence of a progressive award triggering event, the total award further comprises a progressive award associated with that active reel set.

20. The method of claim 19, wherein for each active reel set, the progressive award triggering event occurs in association with that active reel set responsive to a quantity of symbols accumulated, over the plurality of plays of the game, at the plurality of symbol display positions associated with that active reel set.

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